

voyAGER: free web interface for the analysis of age- related gene expression alterations in human tissues

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Abstract

We herein introduce voyAGER, an online graphical interface to explore age-related gene expression alterations in 49 human tissues. voyAGER offers a visualisation and statistical toolkit for the finding and functional exploration of sex- and tissue-specific transcriptomic changes with age. In its conception, we developed a novel bioinformatics pipeline leveraging RNA sequencing data, from the GTEx project, encompassing more than 900 individuals. voyAGER reveals transcriptomic signatures of the known asynchronous ageing between tissues, allowing the observation of tissue-specific age-periods of major transcriptional changes, associated with alterations in different biological pathways, cellular composition, and disease conditions.

Notably, voyAGER was created to assist researchers with no expertise in bioinformatics, providing a supportive framework for elaborating, testing and refining their hypotheses on the molecular nature of human ageing and its association with pathologies, thereby also aiding in the discovery of novel therapeutic targets. voyAGER is freely available at <https://compbio.imm.medicina.ulisboa.pt/app/voyAGER>.

eLife assessment

This work presents an **important** online platform designed to facilitate the exploration of genes and genetic pathways implicated in human aging. Leveraging a new inference methodology, the tool enables the identification and visualization of key genes and tissues impacted by aging, facilitating scientific discovery. The methods and analyses are **convincing** and will be broadly used by scientists aiming to mine human aging RNA-seq data.

Introduction

The ageing-associated progressive loss of proper tissue homeostasis maintenance makes age a prevalent risk factor for many human pathologies, including cancer, neurodegenerative and cardiovascular diseases ^{1–3}. A better comprehension of the molecular mechanisms of human ageing is thus required for the development and effective application of therapies targeting its associated morbidities.

Dynamic transcriptional alterations accompany most physiological processes occurring in human tissues ⁴. Transcriptomic analyses of tissue samples can thus provide snapshots of cellular states therein and insights into how their modifications over time impact tissue physiology. A small proportion of transcripts has indeed been shown to vary with age in tissue- ⁵ and sex-specific ^{6–11} manners. Such variations reflect dysregulations of gene expression that underlie cellular disfunctions ⁵.

Many studies analysed the age-related changes in gene expression in rodent tissues ^{12–18}, emphasising the role in ageing of genes related to inflammatory responses, cell cycle or the electron transport chain. However, while it is possible to monitor the modifications in gene expression over time in those species by sequencing transcriptomes of organs of littermates at different ages, as a close surrogate of longitudinality, such studies cannot be conducted in humans for ethical reasons. Indeed, most studies aimed at profiling ageing-related gene expression changes in human tissues focused on a single tissue (e.g. muscle ^{19–21}, kidney ²², brain ^{23–27}, skin ^{28,29}, blood ^{30,31}, liver ³², retina ³³) and/or are limited to a comparison between young and old individuals ^{20,21,28,32,33}, failing to fully capture the changes of the tissue-specific gene expression landscape throughout ageing ⁵. A few studies were nonetheless led on more than one tissue in humans, from post-mortem samples ^{34,35} and biopsies ^{36,37}, and in mice ^{18,36} and rats ³⁸. The age-related transcriptional profiles derived therein, either from regression ^{18,34,35,37} or comparison between age groups ^{36,38}, highlight an asynchronous ageing of tissues (discussed in ³⁹), with some of them more affected by age-related gene expression changes associated with biological mechanisms known to be impacted by ageing such as mitochondrial activity or metabolic homeostasis. In particular, tissue-specific periods of major transcriptional changes in the fifth and eighth decades of human lifespan have been revealed ³⁴, reflecting the so-called digital ageing ³⁹, consistent with what is observed in mice ^{17,18}. Furthermore, despite outlining the tissue specificity of the transcriptomic signatures of human ageing, some synchronisation was found between tissues like lung, heart and whole blood, which exhibit a co-ageing pattern ³⁵. Nevertheless, as each study followed its own specific procedures, from sample collection to data processing, results from these analyses are hard to compare with one another.

Processed data from those studies have not been made easily accessible and interpretable to researchers lacking computational proficiency but aiming to use them to test their novel hypotheses. To fill this void, we have developed voyAGER, a web application providing flexible visualisation of comprehensive functional analyses of gene expression alterations occurring in 49 human tissues with age in each biological sex. We leverage the large RNA-seq dataset from the Genotype-Tissue Expression (GTEx) project ⁴⁰, encompassing tissue samples from hundreds of donors aged from 20 to 70 years, with a pipeline for gene expression profiling with an optimised temporal resolution. voyAGER allows to investigate ageing from two perspectives: (i) gene-centric – how each gene's tissue-specific expression progresses with age; and (ii) tissue-centric – how tissue-specific transcriptomes change with age. Additionally, voyAGER enables the examination of modules of co-expressed genes altered with age in 4 tissues (brain cortex, skeletal muscle, left ventricle of the heart, whole blood), namely their enrichment in specific cell types, biological pathways, and association with diseases. We therefore expect voyAGER to become a valuable support tool for researchers aiming to uncover the molecular mechanisms underlying human ageing.

Moreover, being open-source, voyAGER can be adapted by fellow developers to be used with alternative datasets (e.g. from other species) or to incorporate other specific functionalities.

voyAGER is freely available at <https://compbio.imm.medicina.ulisboa.pt/app/voyAGER>.

Results

voyAGER's interactive exploration of tissue-specific gene expression landscapes in ageing is based on sequential fitting of linear models (v. Materials and Methods) to estimate, for each gene in each tissue:

1. the *Age* effect, i.e., how the age-associated changes in gene expression evolve with age itself;
2. the *Sex* effect, i.e., how the differences in gene expression between sexes evolve with age;
3. the *Age&Sex* interaction effect, i.e., how the differences between sexes of age-associated changes in gene expression evolve with age.

We named our approach Shifting Age Range Pipeline for Linear Modelling (ShARP-LM). Briefly, this method consists in performing differential gene expression (with gene expression as a function of the donors' *Age*, *Sex* and *Age&Sex* interaction) in moving age windows spanning 16 years. By considering the percentage of genes altered in each age range, we can highlight age periods of major tissue-specific transcriptomic alterations (**Figure 1**).

Gene-centric analyses of human tissue-specific expression changes across age

The progression of tissue-specific expression of a particular gene across age can be examined in voyAGER's *Gene* tab. By entering its HGNC symbol in the Gene selector, the user has access to graphical summaries of the gene's tissue-specific expression (sub-tab *Profile*) (**Figure 2A**) and significance of age-related changes in its expression due to the *Age*, *Sex*, and *Age&Sex* effects (sub-tab *Alteration*) (**Figure 2B**) across age. Results can be displayed in a heatmap for all tissues or in a scatter plot for a chosen individual tissue (**Figure 2C**). When the gene is studied in a single tissue, the user can graphically and statistically profile the association of the donors' sex and reported conditions (e.g., history of heart attack or pneumonia) with the gene's expression profile. A table summarizing the donors' metadata is also shown (**Figure 2C**). The user can interactively select donors of interest on the scatter plot and further examine their information in the automatically subsetting table.

An example of a process whose molecular mechanisms are of particular interest to researchers in the ageing field is cellular senescence. Senescence is a stress-induced cell cycle arrest limiting proliferation of potentially oncogenic cells but progressively creating an inflammatory environment in tissues as they age^{41,42}. *CDKN2A*, that encodes cell cycle regulatory protein p16^{INK4A} known to accumulate in senescent cells^{43,44}, has its expression increased with age in the vast majority of tissues profiled (**Supplementary Figure 1A**).

Similarly, reduced levels of proliferation markers, such as *PCNA*⁴⁵ and *MKI67*⁴⁶, can be studied as putative markers of ageing of certain tissues. These genes have their expression altered with age in the lung and display a similar expression profile (decreasing from 25 to 30 years old (y.o.), constant between 35 and 50 y.o. and decreasing in older ages) (**Figure 2C**). However, these trends appear to vary according to the donor's history of non-metastatic cancer (**Supplementary Figure 1B**), illustrating voyAGER's use in helping to find associations between gene expression and age-related diseases.

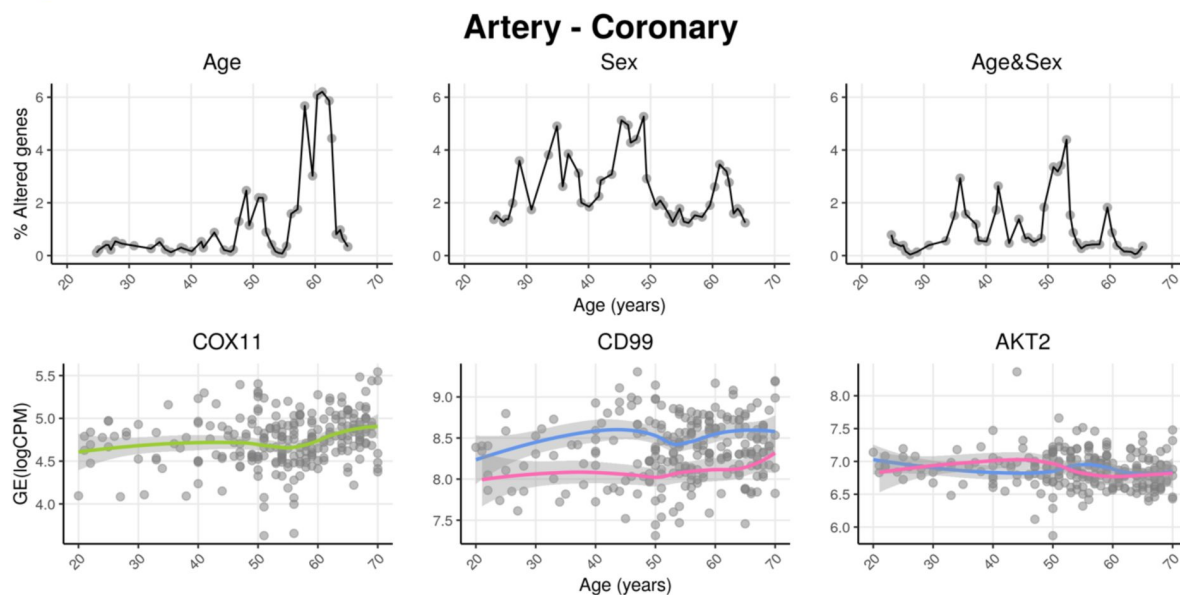


Figure 1

voyAGER profiles tissue-specific age-associated changes in gene expression and their differences between sexes.

For each of the 49 human tissues in GTEx, gene expression was linearly modelled in windows spanning 16 years centred in consecutive years of age, to estimate the effects thereon of *Age*, *Sex* and the interaction between them, i.e., how the *Sex* effect changes with age, equivalent to how the *Age* effect differs between sexes (v. Materials and Methods). In each age window, the percentage of genes with expression significantly altered by of each of those effects gives their respective transcriptomic impact (upper panels). voyAGER thereby identifies the age periods at which major gene expression changes occur in each tissue. For example, in coronary artery: major age-related transcriptional alterations are found at around 60 years of age (upper left panel), illustrated by the behaviour of *COX11* (bottom left panel); major gene expression differences between males and females happen across the considered age range (upper centre panel), as illustrated by *CD99* (bottom centre panel); major differences between sexes in age-related gene expression alterations happen across the considered age range (upper right panel), as illustrated by *AKT2* (bottom right panel). Solid loess lines in the bottom panels (green for all donors, pink for females, blue for males). Gene expression (GE) in $\log_2$ of counts per million (logCPM).

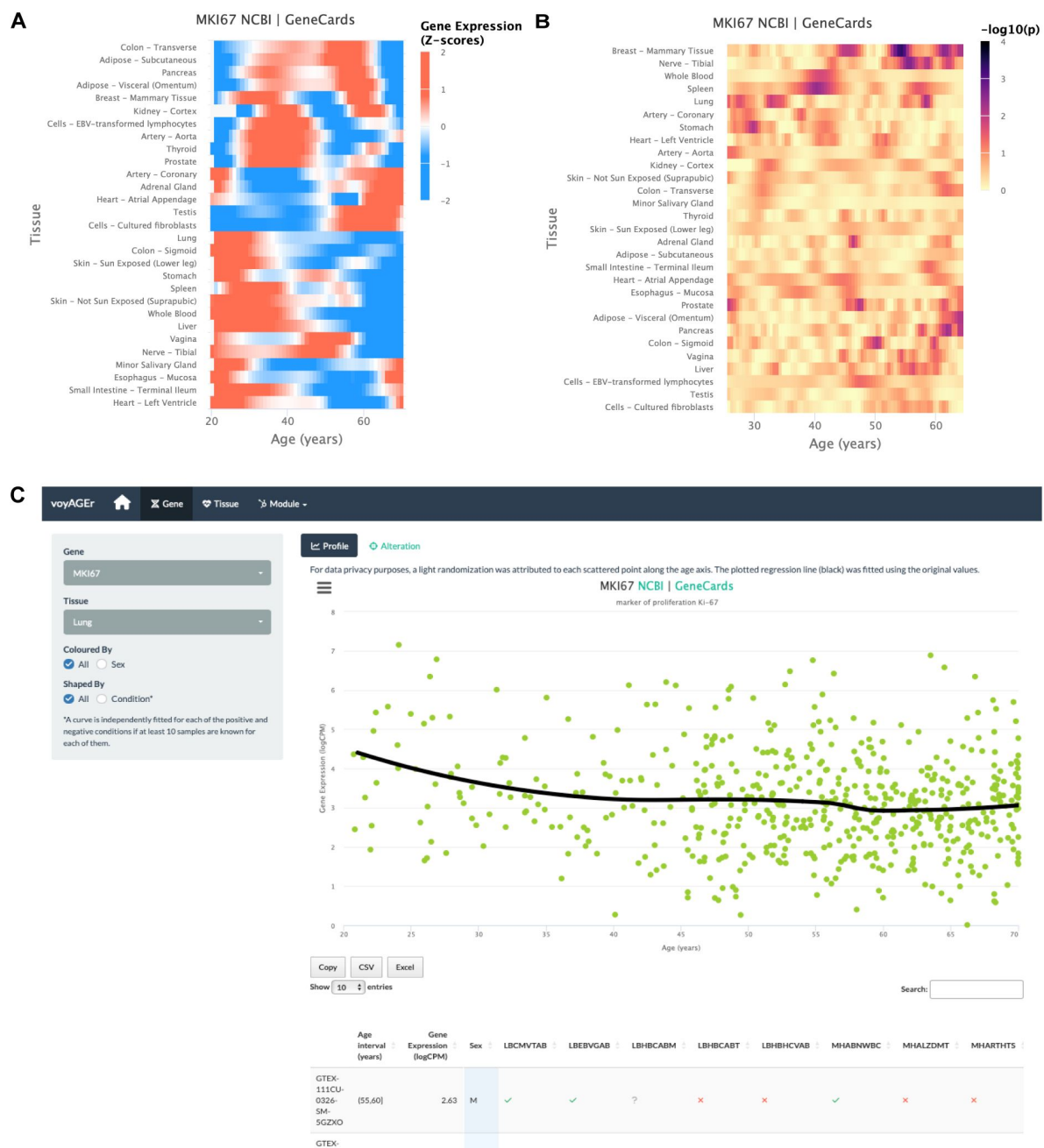


Figure 2

Gene-centric analyses of expression alterations across age

A – Heatmap of *MKI67* expression across tissues over age. **B** – Heatmap of significance of Age-associated *MKI67* expression alterations over age. p-values are for the moderated t-statistics of differential gene expression associated with the Age effect (v. the ShARP-LM approach in Materials and Methods). Notably, transcriptional changes are observed in lung (mid 20's, early 30's and after 55). **C** – voyAGER's *Gene* tab interface. *MKI67* expression in lung is inspected. Donors' information is shown in a table and the scatter plot can be interactively adjusted according to the donors' condition of interest (**Supplementary Figure 1B**).

On a different note, sex biases have been reported in the expression of *SALL1* and *DDX43* in adipose tissue and lung, respectively¹⁰, voyAGER allows to not only recapitulate those observations but also assess the temporal window where these changes occur (**Supplementary Figure 2**).

Tissue-specific assessment of gene expression changes across age

Peaks of gene expression alterations

The landscape of global tissue-specific gene expression alteration across age can be examined in voyAGER's *Tissue* main tab. A heatmap displaying, for all tissues, the statistical significance over age (v. Materials and Methods) of the proportion of genes altered with *Age*, *Sex* or *Age&Sex* (depending on the user's interest) is initially shown (**Figure 3A**), illustrating the aforementioned asynchronous ageing of tissues observed for humans and rodents^{18,32–36,39}.

The user can then spot the age periods with the most significant gene expression alterations in a selected tissue (**Figure 3B a**), and identify the associated altered genes (**Figure 3B b**).

The user can also plot the expression of a given gene of interest across age together with the significance of its expression modification with *Age*, *Sex* or *Age&Sex* (**Figure 3B c**).

An example of voyAGER's capabilities is illustrated in **Supplementary Figure 3**, showing substantial transcriptomic alterations in the uterus from the late forties to the early fifties, overlapping with the age distribution of menopausal onset⁴⁷, which could explain the observed molecular modifications.

It is also possible to visualise tissues with more than one period of transcriptomic changes, and to individually inspect these periods. As an example, the subcutaneous adipose tissue appears to go through two main periods of transcriptional changes with age: a major one at the late 20's (~13 % of altered genes), and a minor one after 45 years (~5% of altered genes) (**Figure 3B a**). The most altered genes in this first peak appear to have their expression modified only at this precise age period (e.g. *PRELID1*, *RUNX1T1*, *TUBB4B*, *FGFR1* and *MALSU1*). Similarly, mitochondrial genes (e.g. *MT-CYB*, *MT-ND4*, *MT-ATP6*, *MT-ND2*) (**Figure 3B b**) appear to be the most altered genes in the second peak (**Figure 3B c**). This particularity suggests that different sets of genes drive the periods of major transcriptional changes, which begs assessing if they reflect the activation of distinct biological processes.

Gene Set Enrichment

The user can explore the biological functions of the set of genes underlying each peak of transcriptomic changes by assessing their enrichment in curated pathways from the Reactome database⁴⁸. voyAGER performs Gene Set Enrichment Analysis (GSEA)⁴⁹ and the user can visualise heatmaps displaying the evolution over age of the resulting normalised enrichment score (NES⁴⁹), reflecting the degree to which a pathway is over- or under-represented in a subset of genes) for a given tissue, effect (*Age*, *Sex* or *Age&Sex*) and Reactome pathway (all, or user-selected) (**Figure 4A a**). To reduce redundancy and facilitate the understanding of their biological relevance, we clustered those pathways into families that also include Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways⁵⁰ and Gene Ontology (GO) Biological Processes of level 3⁵¹. Briefly, we clustered gene sets from the 3 sources based on the overlap of their genes (v. Materials and Methods), thereby creating families of highly functionally related pathways. Taking advantage of the complementary and distinct terminology in Reactome, KEGG and GO, the user can interpret each family's broad biological function by looking at the word cloud of its most prevalent terms, and browsing the list of its associated pathways (**Figure 4A b**). For example, although most pathways enriched in the two aforementioned peaks of altered genes in

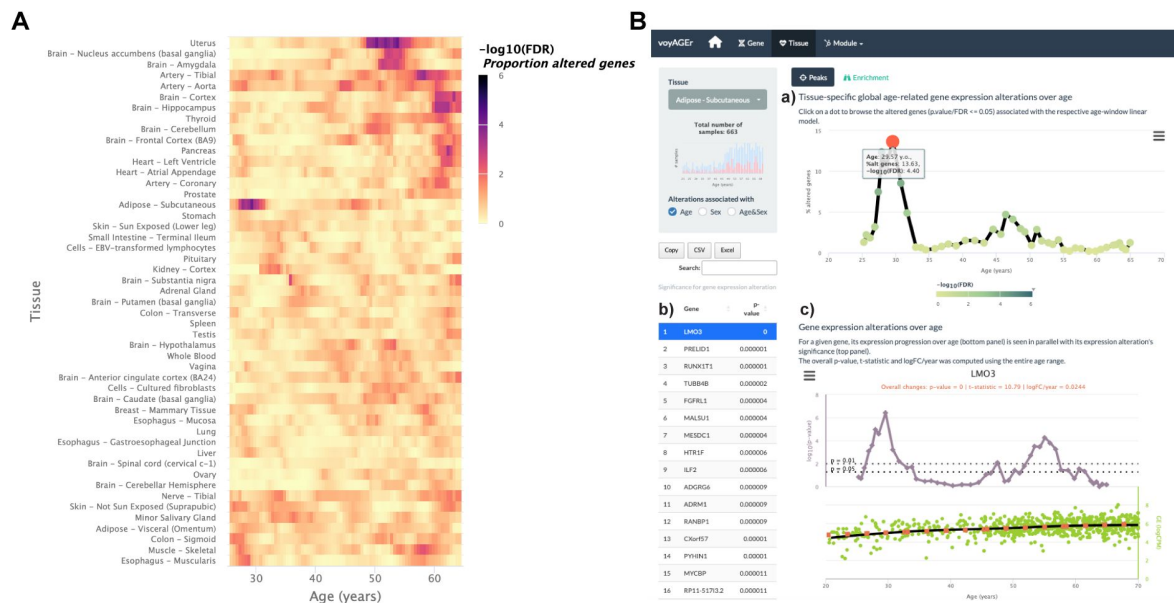


Figure 3

Tissue-specific assessment of gene expression changes across age

A – Heatmap of significance (FDR based on random permutations of age, v. Materials and Methods) over age of the proportion of genes with expression significantly altered with Age in the 49 analysed tissues. **B** – Exploration of gene expression changes across age in Subcutaneous Adipose tissue: (a) Percentage of genes with significantly altered expression with Age over age. Two main peaks of transcriptional changes are noteworthy, a major one at the late 20's and a minor one after 45; (b) Clicking on the dot of a specific age (29.57 y.o. in plot a) gives access to the list of the most altered genes at that age, ordered by statistical significance of expression changes (p-value of moderated t-statistic). (c) Plot of expression of the chosen top gene in the table in b across age (bottom) in parallel with the significance of its expression alterations with Age. The expression of *LMO3* significantly increases at around 30 years old, concomitantly with the first peak of transcriptomic changes with Age.

subcutaneous adipose tissue were different, there is an overlap of pathways related to metabolism, including various mitochondrial processes (**Figure 4A** [↗](#)). This highlights the importance of integrating individual gene data with pathway enrichment analysis to garner more comprehensive insights on the biological relevance of those changes.

The peaks of transcriptomic changes can also be examined for enrichment in a user-provided gene set (**Figure 4B** [↗](#)). As expression of senescence-related *CDKN2A* is increased in the subcutaneous adipose tissue with age (**Supplementary Figure 1A**), we hypothesised that other senescence-associated genes may be augmented too. Thus we used that voyAGER functionality, using the Senequest ⁴² [↗](#) geneset (supported by at least 4 sources) to test it, observing no significant alterations (**Figure 4B** [↗](#)).

Modules of co-expressed genes

voyAGER also allows functional analyses of modules of co-expressed genes (i.e. genes with highly correlated expression across samples, defined by weighted correlation network analysis ⁵² [↗](#)). Genes in the same module are likely to be co-regulated and share biological functions or associations with phenotypical or pathological traits ⁵³ [↗](#). Those modules may also act as markers of core transcriptional features of cellular activity and identity ⁵⁴ [↗](#).

Concretely, voyAGER enables the user to visualise how the expression of modules of genes that are associated with a specific cell type, biological pathway or disease progresses over age in a specific tissue. After selecting one tissue of interest, the user has, for each module, access to four levels of information:

1. *Expression*: its eigengene expression progression over age (**Figure 5C** [↗](#));
2. *Cell types*: its enrichment in specific cell types, based on cell type signatures found in the literature (**Figure 5C** [↗](#));
3. *Pathways*: its enrichment in Reactome pathways;
4. *Diseases*: its enrichment in disease markers, based on gene-disease associations from DisGeNET ⁵⁵ [↗](#), ⁵⁶ [↗](#) calculated with both the disgenet2r package ⁵⁵ [↗](#) and with Fisher's tests (**Figure 5B** [↗](#)).

By default, for each tissue, results are displayed in the form of heatmaps of expression (centred and scaled) illustrating how all modules evolve with age (**Figure 5A** [↗](#)). The user also has the possibility to select a module of interest and see its eigengene progression over age in a scatter plot (**Figure 5C** [↗](#)), lists of its association with diseases and pathways ordered by significance, and a TreeMap for its cell type enrichments (**Figure 5C** [↗](#)). In the example of **Figure 5C** [↗](#), the “MEblue” module, comprising 241 genes co-expressed in the brain cortex, shows significant enrichment in astrocyte markers. The apparent increase of this module's expression with age may reflect the known age-related changes in astrocyte activation ⁵⁷ [↗](#) and the relative weakening of neuronal activity (“MEyellow” and “MEpink” modules).

As in the *Gene* tab, the user can separate donors based on their sex and associated medical conditions in the scatter plot of the eigengene expression progression. On the Pathways and Diseases-Manual tabs below the main plot, the user can also visualise the contingency table from that specific disease/pathway, on the corresponding column.

Discussion

voyAGER provides a framework to examine the progression of gene expression over age in several human tissues, serving as a valuable resource for the ageing research community. In particular, it helps to identify tissue-specific age periods of major transcriptomic alterations. The results of our



Figure 4

Tissue-specific assessment of pathway expression changes across age in the human Subcutaneous Adipose tissue

A – Heatmap depicting the normalised enrichment scores (NES) of Reactome pathways associated with specific tissues and effects. Pathways are classified into 10 families (a), which can be characterised by their frequently occurring terms (b), providing insights into their biological functions. Only pathways significantly linked to gene expression changes in at least one age window (FDR ≤ 0.01) are displayed. Black squares indicate the two age periods with prominent transcriptional changes, while yellow squares denote pathways common to both peaks, primarily belonging to family 2. Word cloud analysis (b) reveals that family 2 pathways are mainly related to metabolism. **B** – Enrichment of a user-provided gene set, given by the significance of Fisher's tests, in genes altered with Age throughout ageing (based on a user-defined p-value threshold). Here, the given gene set is composed of genes from Senequest ⁴² whose link with senescence is supported by at least 4 sources. In this case, there are no significant peaks.

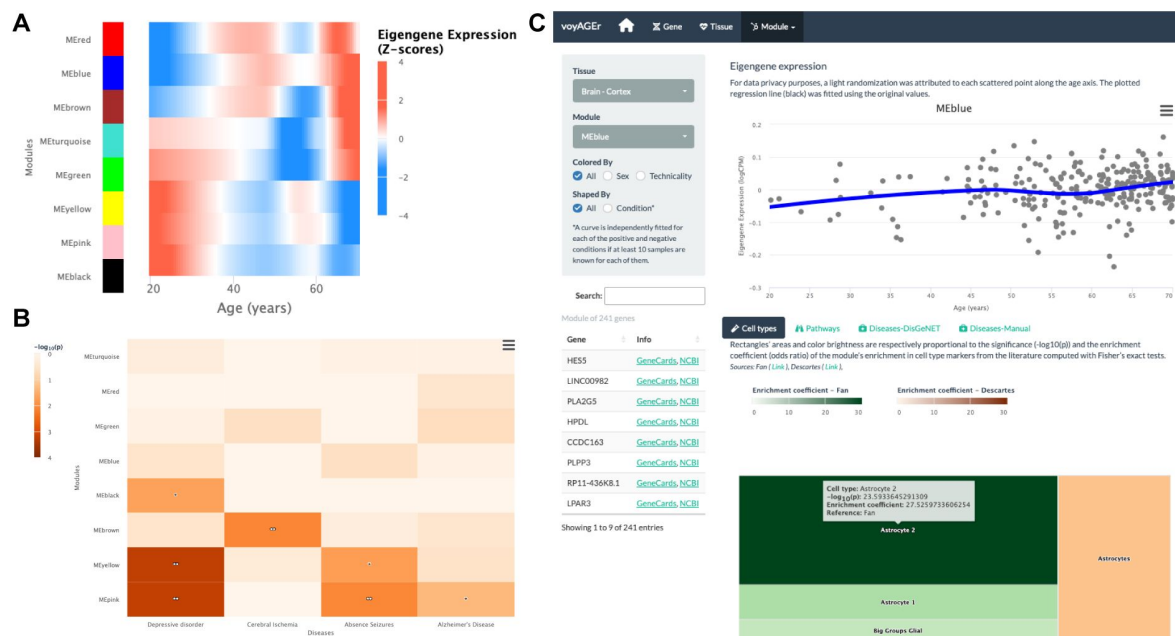


Figure 5

Tissue-specific assessment of age-associated progression of modules of co-expressed genes

A – Heatmap of eigengene expression for all the modules of co-expressed genes in the brain cortex over age. **B** – Heatmap of association of the modules with four selected diseases, computed with the *disgenet2r* package⁵⁵. **C** – Scatter plot (above) of eigengene expression over age, in all brain cortex samples, for a selected module of 241 genes (MEblue). The eigengene expression is derived from the 1st component of the single value decomposition of its genes' expression. This module was analysed, with Fisher's tests, for enrichment in cell types, based on markers from the literature (Fan⁸⁴ and Descartes⁸⁵) and found to be associated with astrocytes, as can be observed by the Treemap below (where each rectangle's area and darkness are proportional to the significance of its association with a cell type and its colour linked to the markers' source study).

analyses show the complexity of human biological ageing by stressing its tissue specificity³⁴ and non-linear transcriptional progression throughout lifetime, consistent with previous results from both proteomic⁵⁸ and transcriptomic^{28,34} analyses. By revealing and annotating the age-specific transcriptional trends in each tissue, voyAGER aims at assisting researchers in deciphering the cellular and molecular mechanisms associated with the age-related physiological decline across the human body.

Due to the tissue-specific nature of the pre-processing steps (v. Read count data pre-processing in the Methods section), and given that most of the plotted gene expression distributions are centred and scaled by tissue, it is important to note that voyAGER may not be always suited for direct comparisons between different tissues. For instance, it does not allow to directly ascertain if a gene exhibits different expression levels in different tissues or if the expression of a particular gene in one tissue changes more drastically with age than in another tissue. Furthermore, we must emphasise that the majority of GTEx donors contributed samples to multiple tissues (**Supplementary Figure 4A**), potentially introducing biases and confounders when comparing gene expression patterns between tissues. Our analyses of variance (**Supplementary Figure 4B**) and downsampling to control for common donors (**Supplementary Figures 4C-E**) suggest very limited global confounding between the impacts of donor and age on gene expression and that any potential cross-tissue bias not to depend much on the proportion of common donors (**Supplementary Figure 4E**). However, this effect must be taken into account when comparing specific pairs of tissues (e.g., Colon – Transverse and Whole Blood, **Supplementary Figure 4D**).

Additionally, voyAGER allows to scrutinise and visually display the tissue-specific differences in gene expression between biological sexes across age. Biological sex is an important factor in the prevalence of ageing-associated diseases, as well as their age of onset, progression^{6,7,59} and sex-related differences in gene expression^{8–11}. By profiling the age distribution of such differences, voyAGER can lead to a better understanding of their influence in the aetiology of the sex specificities of human ageing. For instance, we were able to corroborate findings on the sex-differential transcriptome of adult humans by Gershoni et al.⁹, with voyAGER emphasising its tissue-specificity and allowing to discriminate the ages at which sex-related biases appear to be more prevalent (**Supplementary Figure 5**).

Nonetheless, it is essential to interpret sex chromosome-specific gene results in voyAGER with caution. For instance, we observe elevated expression of Y-chromosome-specific *DDX3Y* in males, whilst its female expression (expected to be zero) is very low, in the range of what can be considered background noise (**Supplementary Figure 6**). Its age-related alterations exhibit a distinctive peak around the age of 40 apparently driven by subtle changes of gene expression in female samples, illustrating the need for the abovementioned caution.

One of the limitations of voyAGER is that most GTEx tissue donors had health conditions and their frequency increased with age, preventing us from defining a class of healthy individuals and identifying age-associated transcriptomic changes that could be more confidently proposed to happen independently of any disease progression. Whilst the large sample sizes and inherent biological variability among individuals, reflected in the diversity of condition combinations, are expected to mitigate significant confounding effects, voyAGER also allows users to evaluate how tissue-specific gene expression trends vary according to the donors' diverse conditions (see **Supplementary Figure 1B**).

The development of voyAGER was accompanied by that of a pipeline, ShARP-LM, that facilitates the holistic depiction of the transcriptional landscapes of adult human tissues throughout ageing with a yearly age resolution. We take advantage of the comprehensiveness of the transcriptome collection from human tissues from the GTEx project to make our analyses a valid surrogate of a currently undoable longitudinal study. It confers our method enough statistical robustness to mitigate the inter-individual differences and deal with the non-uniform distribution of the donors'

ages. Nevertheless, it is worth highlighting that the age distribution of donors does impact the statistical power for detecting transcriptional changes. Consequently, we are more likely to identify significant alterations (with p -value < 0.05 in our gene-centric analyses) within age ranges that are more prevalent in our sample population, often characterised by older individuals (**Supplementary Figure 7**). When downsampling to ensure a balanced age distribution, a loss of statistical power is apparent but a considerable positive correlation with the original results is maintained and a substantial number of significant alterations remain so (**Supplementary Figure 8**). This limitation is likely to be overcome by the accumulation of transcriptomes of human tissues in public databases, promising a gradual increase in accuracy and age resolution with which human transcriptomic ageing can be profiled.

Similarly, the expanding collection of single-cell transcriptomes in public databases is yielding improved gene markers for an increasing diversity of human cell types, enhancing the usefulness of leveraging bulk transcriptomes to study the impact of ageing on the cellular composition of human tissues, for which the co-expression module approach in voyAGER provides a proof-of-concept.

Nonetheless, it is imperative to approach the module-based analysis with caution, as direct and literal interpretations may be misleading. For instance, it is not uncommon to observe an enrichment of “Rheumatoid Arthritis” in modules associated with various immune cell types in anatomical locations, such as the brain cortex, where the disease does not directly manifest. If a specific module associated with a condition like “Liver Cirrhosis” exhibits an age-related increase in the brain cortex, of course this does not mean such disease ever occurs within older brains. Nevertheless, we consider that the module-based approach can serve as a valuable resource for generating hypotheses.

Given its open-source nature, voyAGER is envisaged to be a continually evolving resource, able to accommodate new data and expand its functionalities, namely by incorporating additional tissues into the modules section and integrating perturbagen data for inference of molecular causes underlying observed gene expression alterations and small molecules to target them for therapeutic purposes ^{60,61}.

As an *in silico* approach with no experimental validation for its results, voyAGER is meant to be a discovery tool, supporting biologists in the exploration of a large transcriptomic dataset, thereby generating, refining or preliminary testing hypotheses, laying the groundwork for subsequent experimental research. It can be an entry point for projects aiming at better understanding the tissue- and sex-specific transcriptional alterations underlying human ageing, to be followed by targeted studies focusing on the functional roles of the most promising markers identified therein in the physiology of ageing. Those marker genes can contribute to the development of more robust and cross-tissue gene signatures of ageing ⁶² and the expansion of age-related gene databases ^{63,64}.

Moreover, the observed diverse and asynchronous changes in gene expression between tissues over the human adult life provides potentially relevant information for the design of accurate diagnostic tools and personalised therapies. On one hand, identifying those changes’ association with specific disorders could have a prognostic value by enabling the identification of their onset before clinical symptoms manifest ³⁶. On the other hand, computational screening of databases of genetic and pharmacologically induced human transcriptomic changes could help to infer their molecular causes and uncover candidate drugs to delay these effects ^{60,61,65,66}.

Methods

Development platforms

Data analysis was performed in R (version 4.1.2) and the application developed with R package Shiny⁶⁷, voyAGER's outputs are plots and tables, generated with R packages highcharter⁶⁸ and DT⁶⁹, respectively, that can easily be downloaded in standard formats (png, jpeg, and pdf for the plots; xls and csv for the tables).

voyAGER was deployed using Docker Compose and ShinyProxy 2.6.1 in a Linux virtual machine (64GB RAM, 16 CPU threads and 200GB SSD) running in our institutional computing cluster.

Read count data pre-processing

The matrix with the RNA-seq read counts for each gene in each GTEx v8 sample was downloaded from the project's data portal (<https://www.gtexportal.org/>)⁴⁰. From the 54 tissues available from GTEx v8, five were discarded (kidney medulla, fallopian tube, bladder, ectocervix, endocervix) due to low (<50) numbers of samples.

Read count data for each tissue were then pre-processed separately. We started by filtering out genes deemed uninformative due to their very low expression across samples: only genes with at least 1 CPM in at least 40% of the samples were kept for analysis (the number of genes analysed for each tissue can be found in **Supplementary Table 1**)⁷⁰. Read counts for those kept genes were used to calculate normalisation factors to scale the raw library sizes, using function `calcNormFactors` from edgeR⁷⁰ that implements the trimmed mean of M-values⁷¹. Read counts were subsequently normalised and log-transformed with the voom function⁷² from package limma⁷³.

However, it is well-established that batch effects, which may stem from variations in sample treatment prior to RNA-seq library preparation, can introduce spurious gene expression differences between samples and result in confounding factors⁷⁴. We therefore conducted an impartial and systematic search for potential batch effects. Firstly, we performed a principal component analysis of gene expression for each tissue, using the `prcomp` package.

We quantified the relation between each condition associated with every sample¹ and the first two principal components, by computing Spearman correlations (for numerical conditions), t-tests (for binary categorical conditions), or analysis of variance (ANOVA) tests (for variables with more than two possible values, and, in the case of numerical variables, fewer than 15 unique values). Conditions that surpassed defined empirical thresholds (p-value < 0.05, Spearman correlation > 0.3, t-test > 10, and ANOVA > 20) were flagged as potential batch effects. Except in brain tissues, the COHORT variable (i.e., whether the samples were collected from organ donors or *post mortem*) appeared to be the main batch effect, with ripple effects on numerous other related conditions (**Supplementary Figure 9**). Moreover, SMRIN (sample's RNA integrity number), DTHHRDY (death classification based on the 4-point Hardy Scale), and MHSMKYRS (smoke years) consistently emerged as conditions associated with the primary axes of variance. The number of genes detected in each sample, determined by the filtration step described above, was also identified as a significant contributor to the primary data variance. We therefore corrected for these five conditions, on a tissue-by-tissue basis, by adapting the `removeBatchEffect` function from the limma package⁷³. Specifically, we employed linear models to estimate the contributions to gene expression of each of those factors and subtracted such contributions from the original logCPM matrix. To ensure biological interpretability of the results, we offset the resulting values to the minimum value in the non-corrected matrix. To prevent sample loss due to missing values for the aforementioned five conditions and since the number of missing values was relatively low compared to the total number of samples, imputation was carried out using the mice package⁷⁵.

The resulting matrix of logCPM-corrected values was used for all downstream analyses. As an illustrative example of the importance of batch removal, the expression of surfactant factor *SFTPA2* was found to be associated with donors on a ventilator⁷⁶. Without batch correction, *SFTPA2* expression would have been associated with age due to higher prevalence of such cases among older individuals (**Supplementary Figure 10**).

ShARP-LM

To model the changes in gene expression with age, we developed the Shifting Age Range Pipeline for Linear Modelling (ShARP-LM). For each tissue, we fitted linear models to the gene expression of samples from donors with ages within windows with a range of 16 years shifted through consecutive years of age (i.e. in a sliding window with window size = 16 and step size = 1 years of age). This was the minimum age span needed to guarantee the presence of more than one sample per window, across all considered tissues. As samples at the ends of the dataset's age range would be thereby involved in fewer linear models, we made the window size gradually increase from 11 to 16 years when starting from the “youngest” samples and decrease from 16 to 11 years when reaching the “oldest” (**Supplementary Figure 11**).

Function `lm` from `limma` was used to fit the following linear model for gene expression (GE):

$$GE \sim E_0 + \alpha.Age + \beta.Sex + \chi.Age\&Sex + \varepsilon$$

For each gene, α , β and χ are the coefficients to be estimated for their respective hypothesised effects. For each sample, *Age* in years and *Sex* in binary were centred and *Age&Sex* interaction given by their product. The coefficients E_0 and χ are thus the expression of the average sample (i.e., with average sex and average age) and the error term.

For each gene in each model (i.e., each age window in each tissue), we retrieved the t-statistics of differential expression associated with the three relevant variables and their respective p-values. We considered the average age of the samples' donors within the age window as the representative age of the observed expression changes.

In summary, for a given tissue and variable (*Age*, *Sex* and *Age&Sex*), ShARP-LM yields t-statistics and p-values over age for all genes, reflecting the magnitude and significance of the changes in their expression throughout adult life.

Gene-centric visualisation of tissue-specific expression changes across age

For visualisation purposes, the trend of each gene's expression progression over age in each tissue was derived through Local Polynomial Regression Fitting, using R function `loess` on logCPM values (**Figures 1**, **2C**, **3B c**). For summarizing in a heatmap a given gene's expression across age in multiple tissues (**Figure 2A**) or the expression of several genes across age in a specific tissue, each gene's regression values are centred and scaled, using R function `scale`.

For summarizing in a heatmap the significance of a given gene's expression changes over age in multiple tissues (**Figure 2B**), cubic smoothing splines are fitted to $-\log_{10}(p)$, with p being the t-statistic's p-value, with R function `smooth.spline`.

Tissue-specific quantification of global transcriptomic alterations across age

To assess the global transcriptomic impact of each of the three modelled effects in each tissue across age, we analysed the progression over age (i.e. over consecutive linear models) of the percentage of genes whose expression is significantly altered (t-statistic's p-value ≤ 0.01) by each effect (**Figure 3B** [↗](#)). To evaluate the significance of each percentage and assess if high percentages can be confidently associated with major transcriptomic alterations, we controlled for their false discovery rate (FDR) by randomly permuting the samples' ages and sexes within each age window fifty thousand times and performing ShARP-LM on each randomised dataset. We were then able to associate an FDR to each percentage of altered genes by comparing it with the distribution of those randomly generated (**Figure 3B** [↗](#) a).

We also applied a linear model across the entire age range, thereby providing users with more insight and supporting evidence into how a specific gene changes with age. For visualisation purposes, we incorporated a dashed orange line, with the logFC per year for the Age effect as slope, in the respective scatter plots (**Figure 3B** [↗](#) c). We depict the Sex effect therein by prominent dots on the average samples, with pink and blue denoting females and males, respectively.

Gene Set Enrichment Analysis (GSEA)

For each *Peak* of significant gene expression modifications, we performed GSEA [49](#) [↗](#) on the ordered (from the most positive to the most negative) t-statistics of differential expression for the respective tissue and age, using R package fgsea [77](#) [↗](#) and the Reactome database [48](#) [↗](#). We extracted the GSEA normalised enrichment score (NES), which represents the degree to which a certain gene set is overrepresented at the extreme ends of the ranked list of genes. A positive NES corresponds to the gene set's overrepresentation amongst up-regulated genes within the age window, whereas a negative NES signifies its overrepresentation amongst down-regulated genes. The NES for each pathway was used in subsequent analyses as a metric of its up- or down-regulation in the *Peak*. The resulting normalised enrichment score (NES) for each pathway was used in subsequent analyses as a metric of its over- or down-representation in the *Peak*.

To optimise computational efficiency and minimise redundancy in the analysed pathways, we only considered pathways containing a minimum of 15 genes and up to 500 genes, as suggested in the GSEA User Guide [78](#) [↗](#). For the sake of clarity in voyAGER's visual representations, we only included pathways with a p-adjusted value less than or equal to 0.05, further narrowing it down to pathways within the top 1% of p-adjusted values.

Additionally, we exclusively featured pathways with at least one significant age *Peak* (FDR ≤ 0.05), as illustrated in **Figure 4A** [↗](#).

Families of pathways

To reduce pathway redundancy and facilitate the assessment of their biological relevance in results' interpretation, we created an unifying representation of pathways from Reactome [48](#) [↗](#), KEGG [50](#) [↗](#), and level 3 Biological Processes from GO [51](#) [↗](#), by adapting a published pathway clustering approach [79](#) [↗](#) to integrate them into families.

The approach relies on the definition of a hierarchy of pathways based on the numbers of genes they have in common. For each two pathways P_i and P_j , respectively containing sets of genes G_i and G_j , we computed their overlap index (OI) [79](#) [↗](#), defined as follows:

$$OI_{ij} = |G_i \cap G_j| / \min(|G_i|, |G_j|)$$

Where $|G_i|$ is the number of genes in set G_i and $|G_i \cap G_j|$ is the number of genes in common between G_i and G_j . $OI_{ij} = 1$ would therefore indicate that P_i and P_j are identical in gene composition or that one is a subset of the other. On the contrary, $OI_{ij} = 0$ would mean that P_i and P_j have no genes in common. To ease the computational work, we removed from the analysis pathways that are subsets of larger pathways (i.e. each pathway whose genes are all present in another pathway).

From the OI_{ij} matrix, from which each row is a vector with the gene overlaps of pathway i with each of the pathways, we computed the matrix of Pearson's correlation between all pathways' overlap indexes with R function `cor`. That matrix was finally transformed into Euclidean distances with R function `dist`, allowing for pathways to be subsequently clustered with the complete linkage method with R function `hclust`. The final dendrogram was empirically cut in 10 clusters (Supplementary Figure 12). Pathways that were initially excluded from the computation for being subsets of others were added to the clusters of their respective *parent* pathways. Each *daughter* pathway with more than one parent was assigned to the cluster of the parent with the smallest number of genes, thereby maximizing the daughter-parent similarity. The `data.table` R package, for fast handling of large matrices, was used in this analysis⁸⁰.

Gene co-expression modules

Gene co-expression modules were defined with R package WGCNA⁵². For each considered tissue, the process began with the identification of a set of informative genes that exhibit high variability across samples (referred to as variable A), thus improving module definition. Next, we calculated the bicorrelation matrix for the expression of all selected genes with the `bicor` function. We then applied soft thresholding by raising all correlation values to the power of β , accentuating stronger correlations. The value of $\beta = 12$ was chosen in accordance to the WGCNA FAQs⁸¹, and after confirmation of a free-scale topology using the `pickSoftThreshold` function. We generated the dissimilarity matrix by subtracting the output of the `TOMsimilarity` function from 1. Gene co-expression modules were then defined using a static tree-cutting algorithm, implemented via the `cutreeStaticColor` function, requiring as parameters a minimum number of genes per module (referred to as variable B) and the tree-cutting height (variable C).

The empirical determination of parameters A to C was guided by the following principles: (i) maximizing cell type signature enrichment, (ii) minimizing the number of genes per module, and (iii) ensuring that the modules' eigengenes exhibited age-related variability. Different combinations of these variables were exhaustively tested. To maintain biological relevance, modules consisting of non-assigned genes or genes lacking substantial supporting evidence, such as pseudogenes, were excluded.

Maximizing cell type enrichment in the modules, with a focus on known markers for specific cell types, has previously been proven successful in unveiling core transcriptional features of cell types in the human central nervous system⁵³. Cell type enrichment analysis relied on Fisher tests, providing odds enrichment scores and significance values (p-values). This involved comparing module genes with the signature, considering as background of genes those included in the module definition for each tissue. We prioritised modules of genes for which we obtained at least one significant result for each cell type (odds ratio (OR) > 1 and p-value < 0.05). The cell type signatures employed in this analysis were sourced from MSigDB's C8 collection⁴⁹, and in the case of Whole Blood, additionally from LM22⁸².

Specific variance thresholds (variable A) were employed: 0.5, 0.4, 0.35, and 0.9 for Brain – Cortex, Muscle – Skeletal, Heart – Left Ventricle, and Whole Blood, respectively. The minimum number of genes per module (variable B) was set at 15, 20, 20, and 15, respectively. Tree-cutting heights (variable C) of 0.95, 0.98, 0.99, and 0.97 were respectively used.

Each module is characterised by a set of genes and an eigengene, represented by the first principal component obtained through singular-value decomposition of the module's gene expressions. Subsequently, voyAGER facilitates an evaluation of cell type enrichment, as described earlier, and enrichment in biological pathways and diseases. The enrichment of modules in cell types, Reactome pathways, and diseases (extracted from DisGeNET ⁵⁵[55](#), ⁵⁶[56](#) database version 7.0) was quantified using Fisher's tests. For disease enrichment, the function `disease_enrichment` from the `disgenet2r` package⁵⁵[55](#) was employed, utilising their curated set of diseases. The significance of these enrichments was determined through p-value/FDR adjustment using Benjamini-Hochberg correction. For visual clarity, only pathways and diseases displaying significant enrichment ($p \leq 0.05$) in at least one module were considered.

The four tissues (Brain – Cortex, Muscle – Skeletal, Heart – Left Ventricle, and Whole Blood) covered by the Module section of voyAGER were selected due to their relatively high sample sizes and availability of comprehensive cell type signatures. The increasing availability of human tissue scRNA-seq datasets (e.g., through the Human Cell Atlas⁸³[83](#)) will allow future updates of voyAGER to encompass a wider range of tissues.

Data and Code availability

Processed GTEx v8 RNA-seq data (read count tables) were downloaded from the project's data portal (<https://www.gtexportal.org/>[57](#)). Donor metadata were obtained from dbGaP – database of Genotypes and Phenotypes (Accession phs000424.v9.p2 project ID 13661). voyAGER's output tables can be directly downloaded in standard xls and csv formats.

The complete source code for voyAGER, including pre-processing and Shiny app, can be accessed on GitHub at the following repository: <https://github.com/DiseaseTranscriptomicsLab/voyAGER>[58](#).

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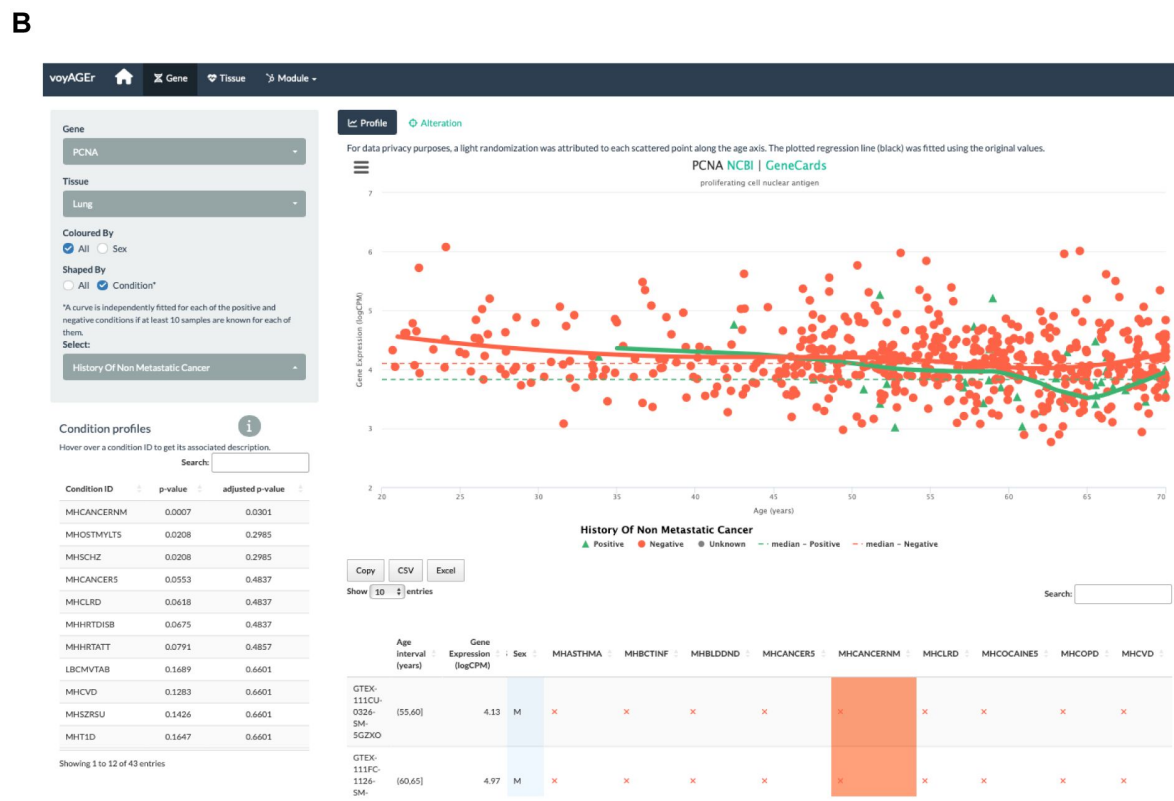
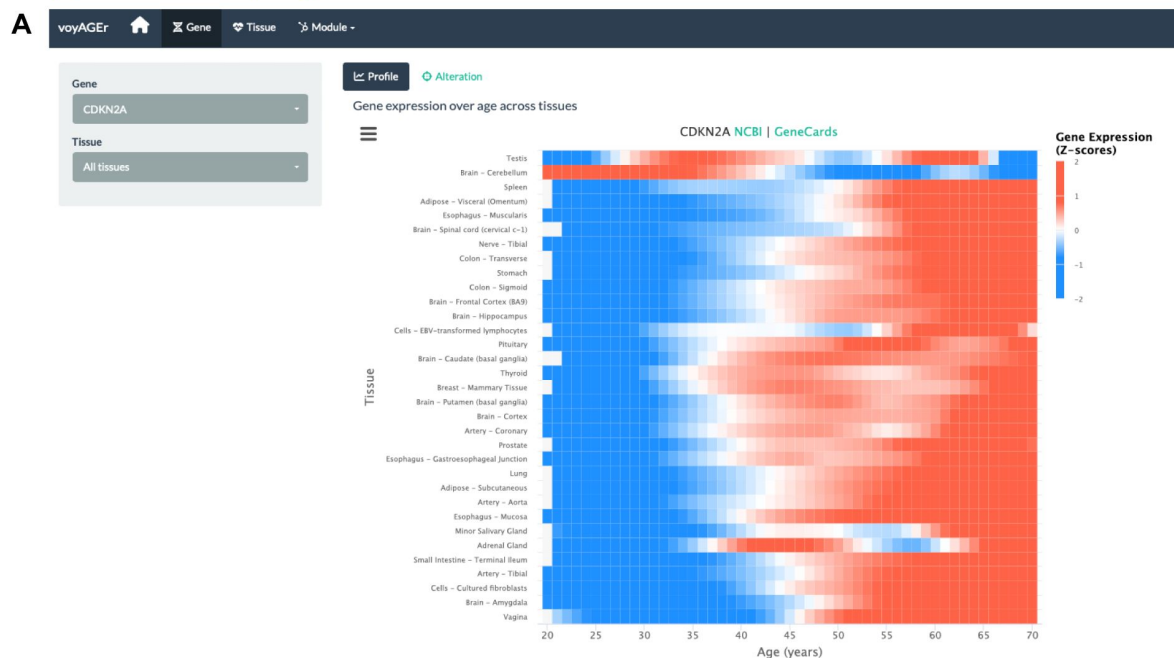
Authors' contributions

A.S. and N.L.B.-M. designed the study, with contributions from R.M.-S. and A.K.; A.S., R.M.-S. and A.K. performed all the development and bioinformatics work, under the supervision of N.L.B.-M.; N.S.-A. set up the web and database implementation; A.S., R.M.-S., A.K., and N.L.B.-M. wrote the manuscript, with contributions from N.S.-A..

Conflict of Interest statement

The authors declare no conflict of interest.

Supplementary Figures and Tables

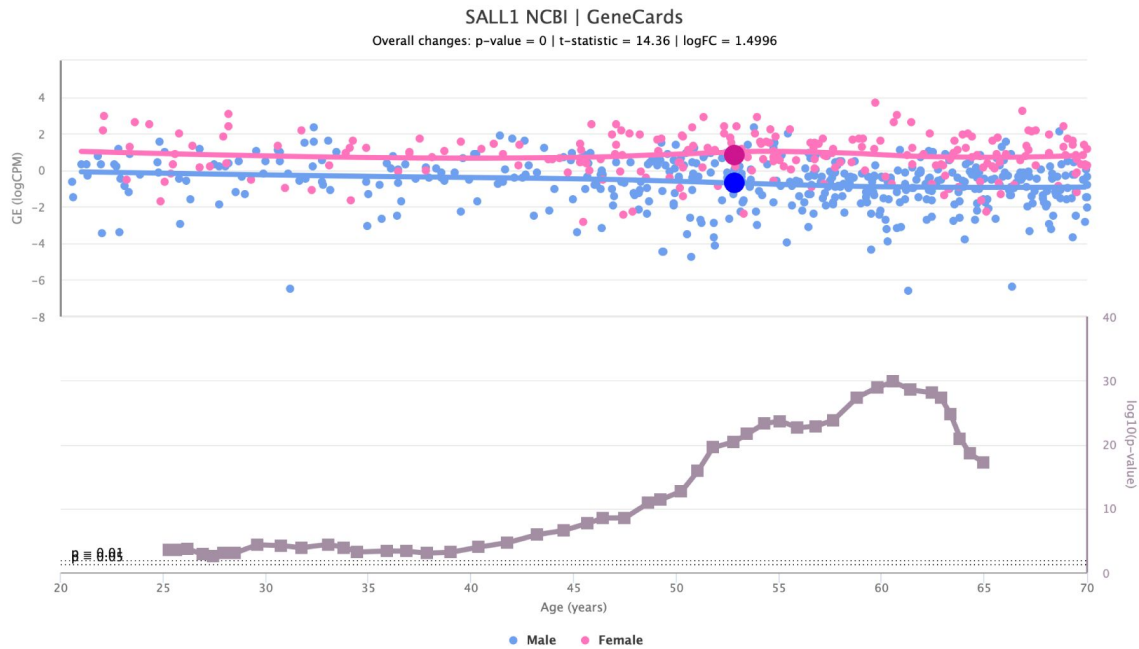


Supplementary Figure 1

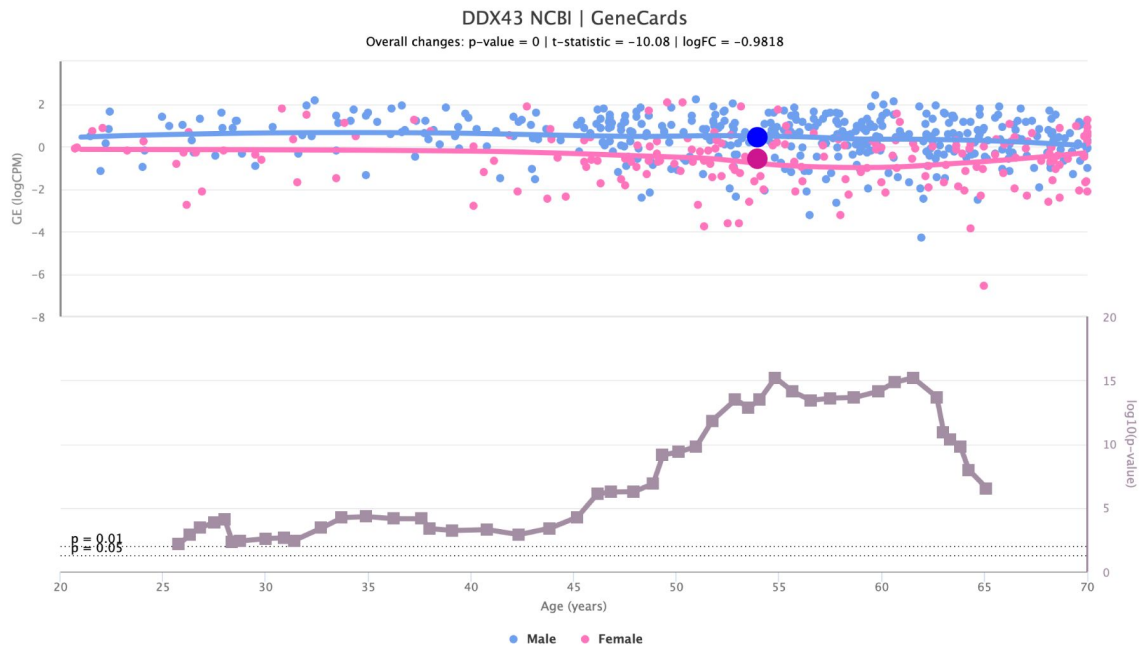
voyAGER's Gene tab interface.

A – Heatmap of *CDKN2A* expression across tissues over age. **B** – *PCNA* expression in lung over age and its association with the donor's history of non-metastatic cancer.

A



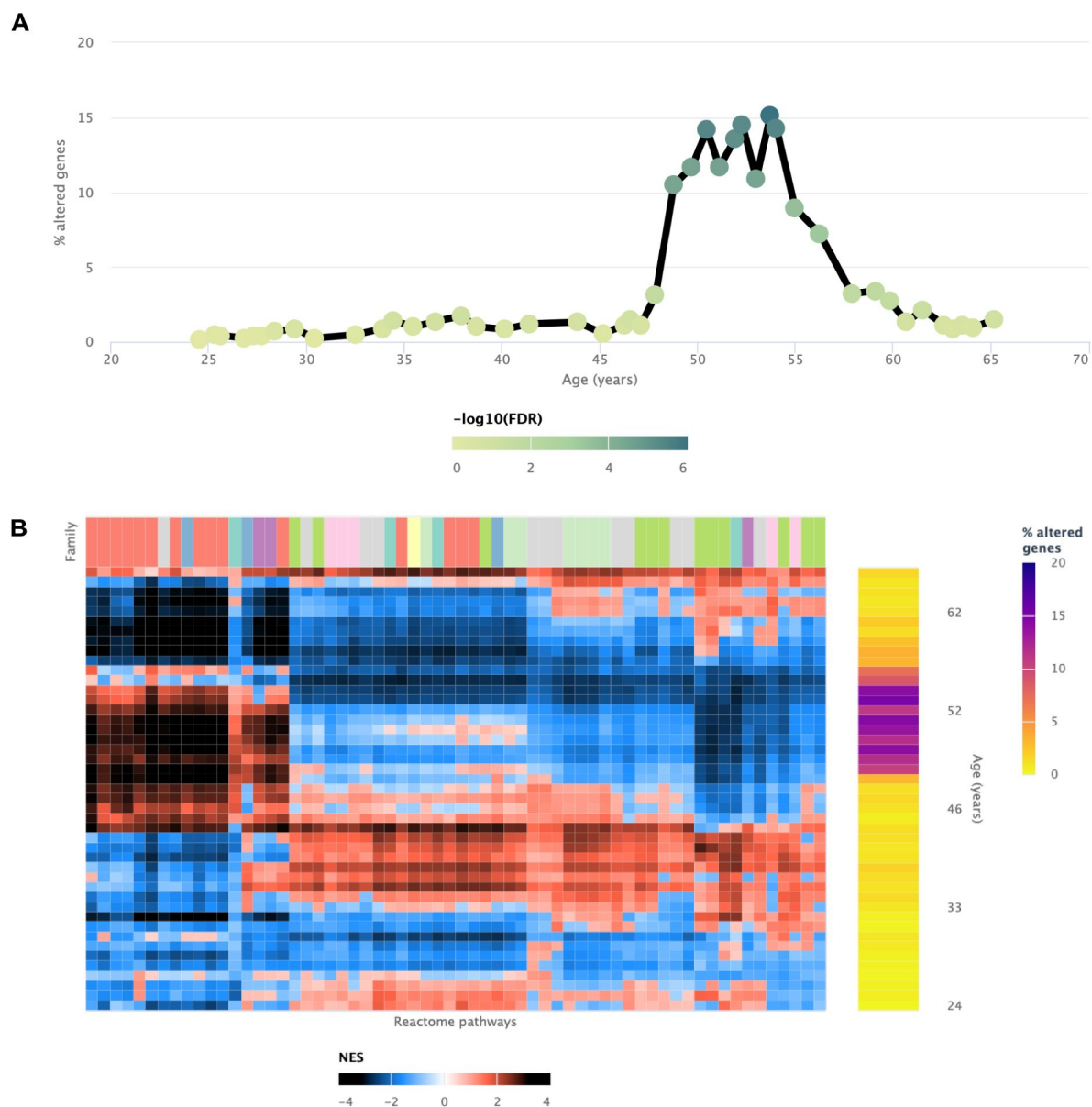
B



Supplementary Figure 2

Sex-specific *SALL1* and *DDX43* expression alterations over age.

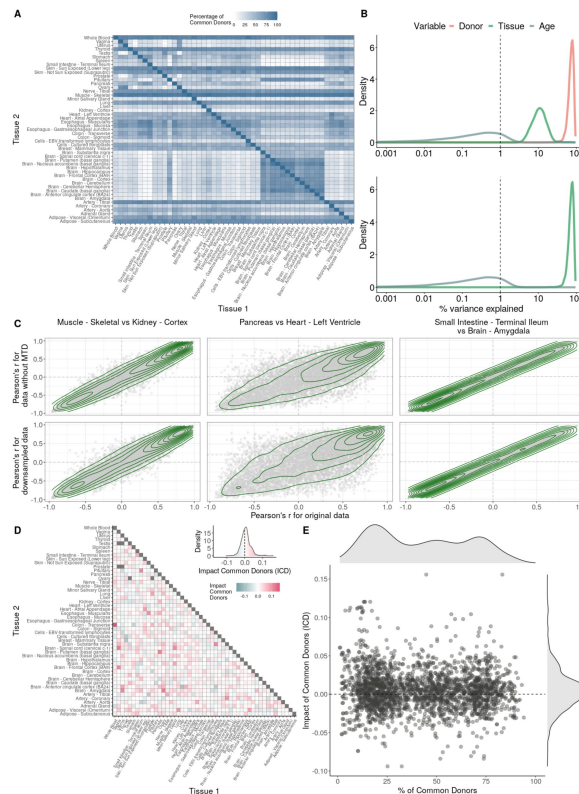
Scatter plots of *SALL1* (**A**) and *DDX43* (**B**) expression over age (above), in adipose tissue and lung, respectively, coloured based on the donor's biological sex (male in blue and female in pink). The scatter plots of gene expression are shown together with line plots of the significance of their difference between sexes (below).



Supplementary Figure 3

Transcriptomic alterations in the uterus coincident with the onset of menopause.

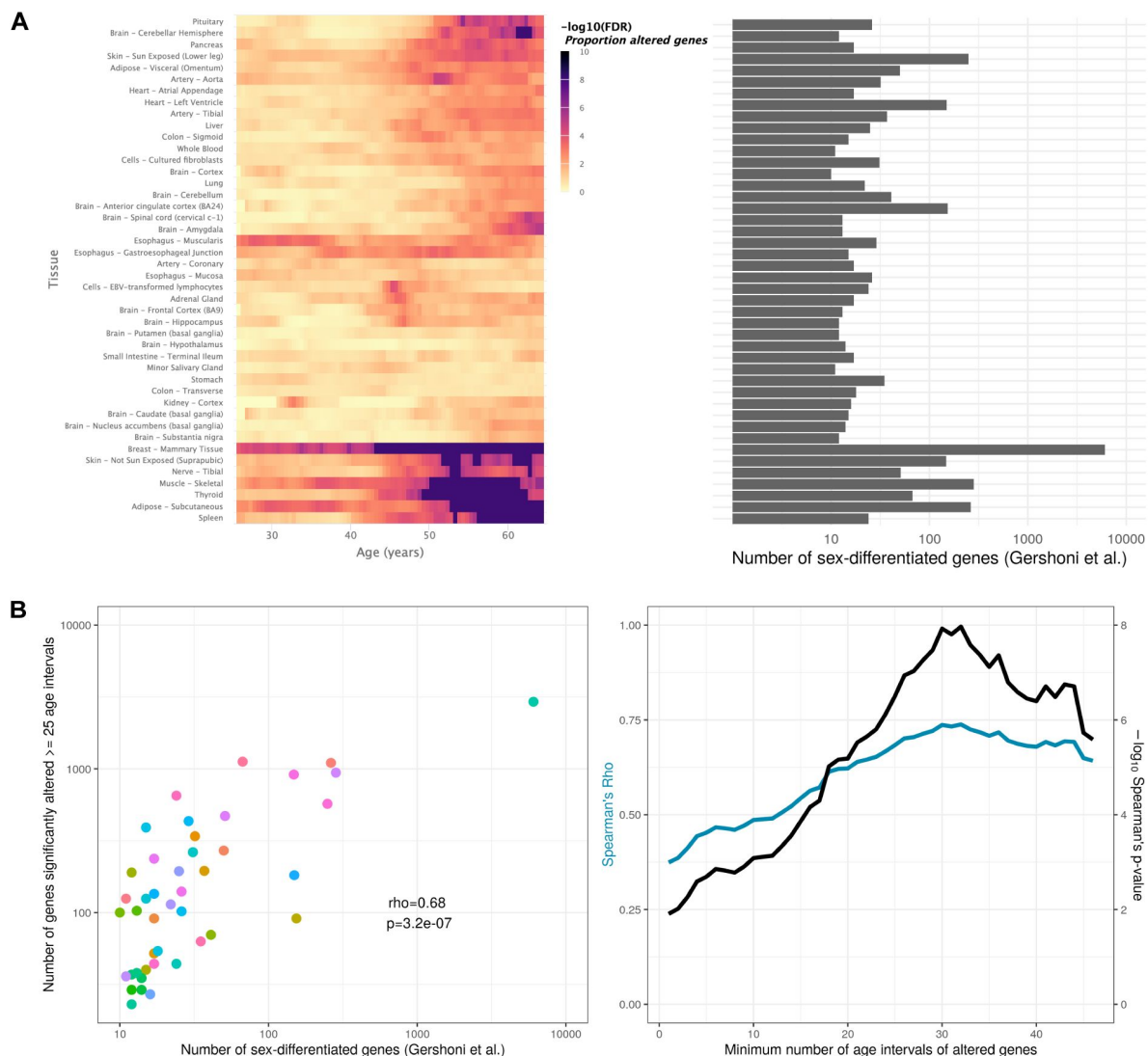
voyAGER enables the identification of a prominent age period of transcriptional alterations in the uterus, which coincides with the onset of menopause⁴⁷. The genes exhibiting significant changes (**A**) reflect the transformations observed in biological processes (**B**), marked by a distinct switch in enrichment scores. Positively enriched pathways become negative, and vice-versa during this age frame.



Supplementary Figure 4

Impact of donor overlap between tissues in age-associated trends

A – Heatmap depicting the percentage of common donors between pairs of tissues. A given square illustrates the percentage of all samples of tissue in the x axis (Tissue 1) that is in common with the tissue in the y axis (Tissue 2). **B** – Assessment of the relative contributions of different sources to the dataset's variance. (Left panel) Tissue accounts for approximately 90% of the total variance, while donor contributes around 10%; age has a minimal impact (1%), likely due to the relative subtlety of its effects on gene expression and to the tissue specificity of ageing dynamics. (Right panel) Removal of the donor variable does not transfer variance to age, suggesting limited confounding between the two variables. **C** – Impact of the relative proportion of common donors on gene expression correlation between tissue pairs. Panels A, B, and C showcase the tissue pairs with the highest (Muscle Skeletal / Kidney Cortex), median (Pancreas / Heart Left Ventricle), and lowest (Small Intestine / Brain Amygdala) percentages of common donors, respectively. The left panels illustrate gene-by-gene Pearson's correlations of gene expression between the two tissues, comparing the scenarios with (x-axis) and without (y-axis) the removal of common donors. The right panels depict the same comparisons, but with random downsampling (y-axis) in both tissues based on the proportions defined for common donor removal. The depicted examples show that the outcomes are comparable when removing common donors or employing random downsampling. **D** – Comparison of the impacts of removing common donor samples and random downsampling across tissue pairs. The process of removing common donors involved the identification and removal of samples from shared donors while maintaining the original sample size imbalance between tissues. As this process inherently involves downsampling, which may impact results, we performed additional downsampling by randomly removing samples from both tissues according to the proportions defined for the removal of common donors. The heatmap is coloured based on whether the removal of common donors has a greater (red) or lesser impact (blue) than random downsampling. The values depicted in the heatmap, denoted as the Impact of Common Donors (ICD), are computed for each tissue pair. This calculation involves several steps: first, by determining the absolute difference in Pearson's correlation for each gene's mean expression within each age window from the ShARP-LM pipeline, between the original data and the subset of data without common donors (DiffWoCD) or with random downsampling (DiffRD). Subsequently, the medians of DiffWoCD and DiffRD are computed, and the difference between these median values provides the ICD for each tissue pair. Due to the unidirectional nature of correlation (i.e., the results for tissue 1 vs tissue 2 mirror those for tissue 2 vs tissue 1), the resulting matrix is triangular in form. Grey tiles denote NA values, i.e., where the tissue-tissue comparison does not have a meaning, namely self-self and between sex-specific tissues. Top right insert: density line ("smoothed" histogram) of all ICD values. **E** – Scatter plot relating the Impact of Common Donors (ICD, values in heatmap D) with the respective percentage of common donors (values in heatmap A).

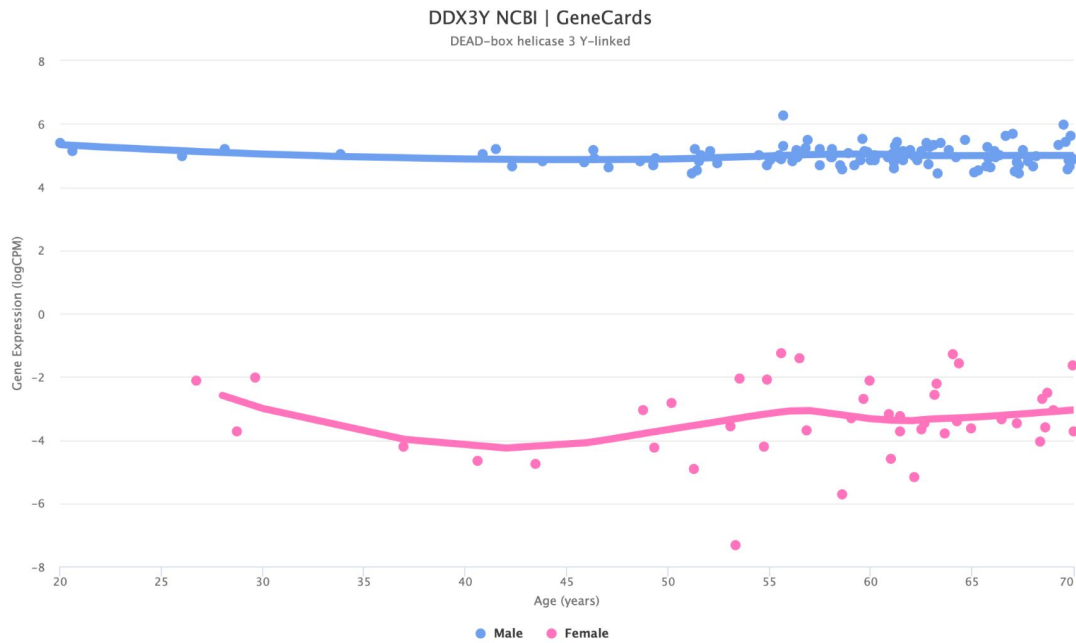


Supplementary Figure 5

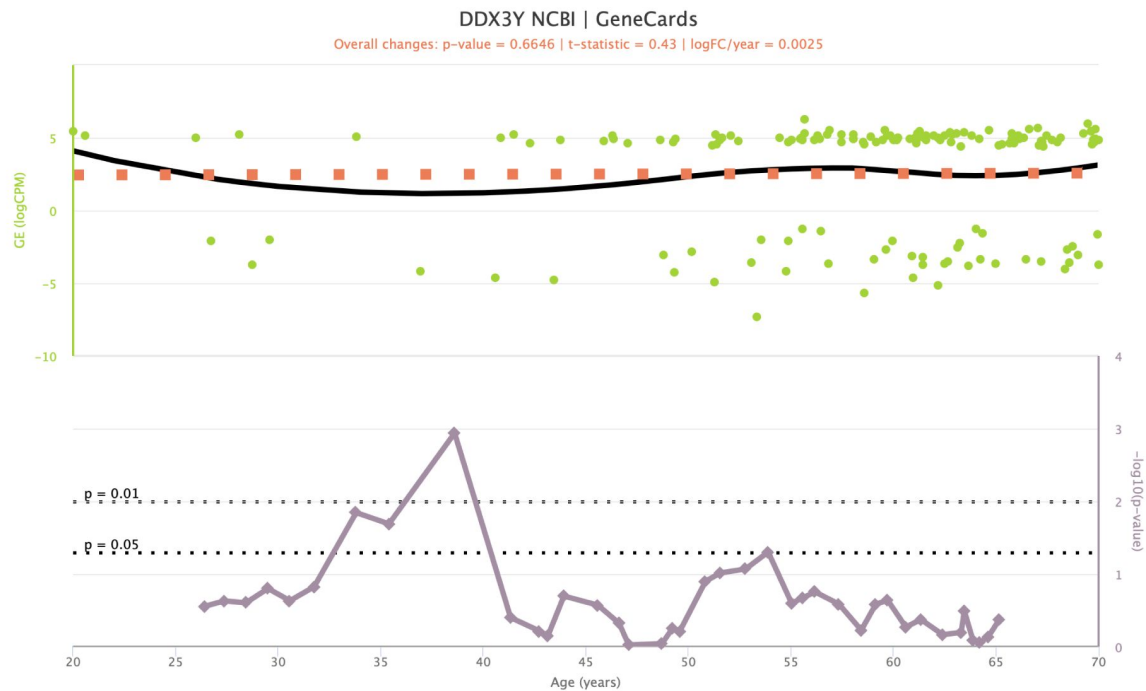
Tissue-specific sex-differentiated expression.

A – Heatmap of significance over age, for each tissue, of the proportion of altered genes between sexes (left), in parallel with the number of sex-differentiated genes highlighted by Gershoni et al. (v. [Figure 1](#) in [9](#)) (right). Mammmary is corroborated as the tissue with the most sex-differentiated gene expression. For tissues like Adipose – Subcutaneous, Skin – Not Sun Exposed, Thyroid, Nerve – Tibial, Muscle – Skeletal, and Spleen, sex-related biases in expression mainly arise in late life. **B** – Scatter plot relating, for each tissue (each dot), the number of sex-differentiated genes found by Gershoni et al. with the number of genes found by voyAGER to be altered ($p\text{-value} \leq 0.05$) between sexes in at least 25 age intervals (left). This correlation is significant ($p\text{-value} \leq 0.05$) regardless of the considered minimum number of age intervals a gene must be altered in (right).

A



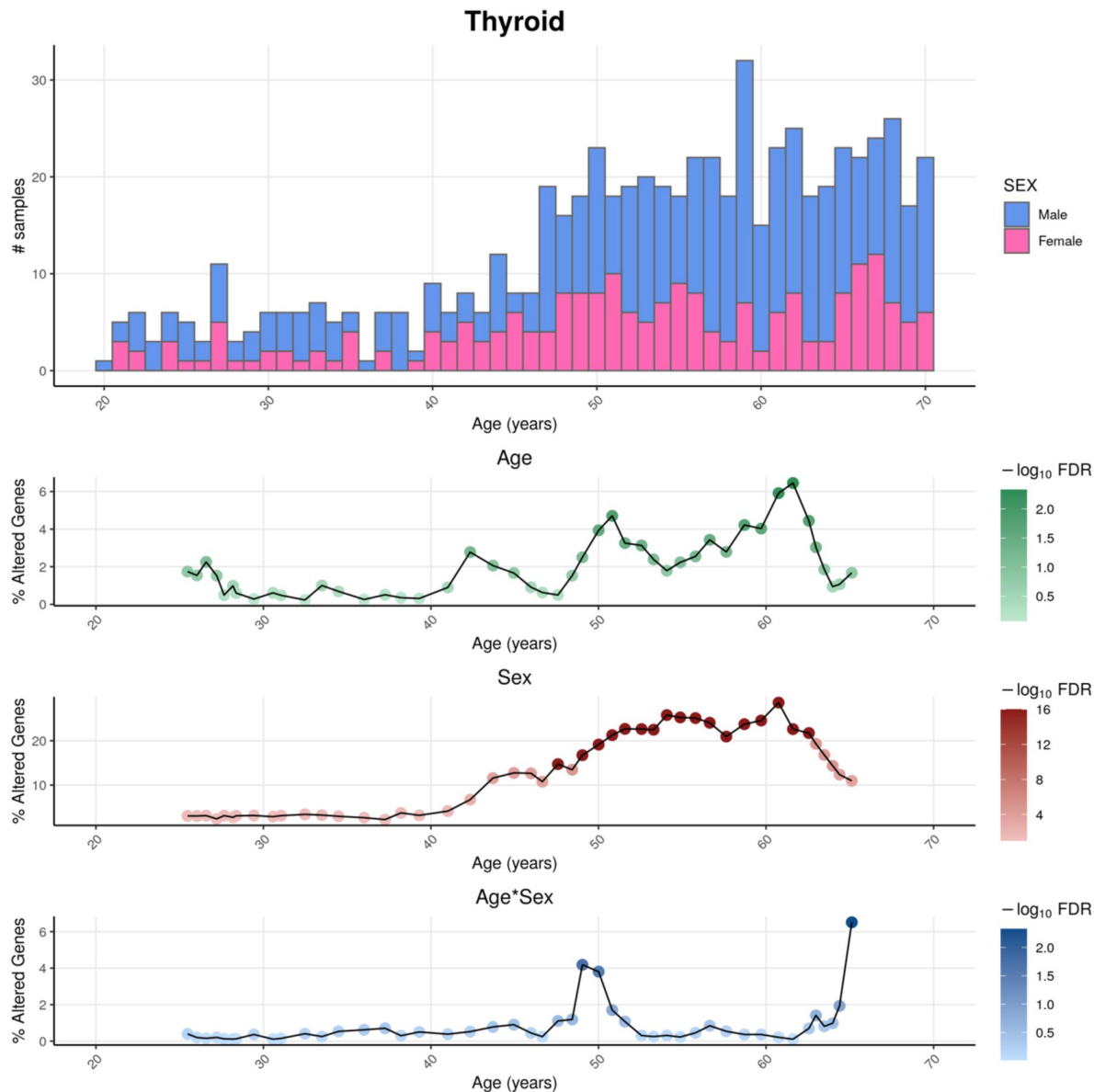
B



Supplementary Figure 6

Impact of sex-specific genes in the interpretation of voyAGER results.

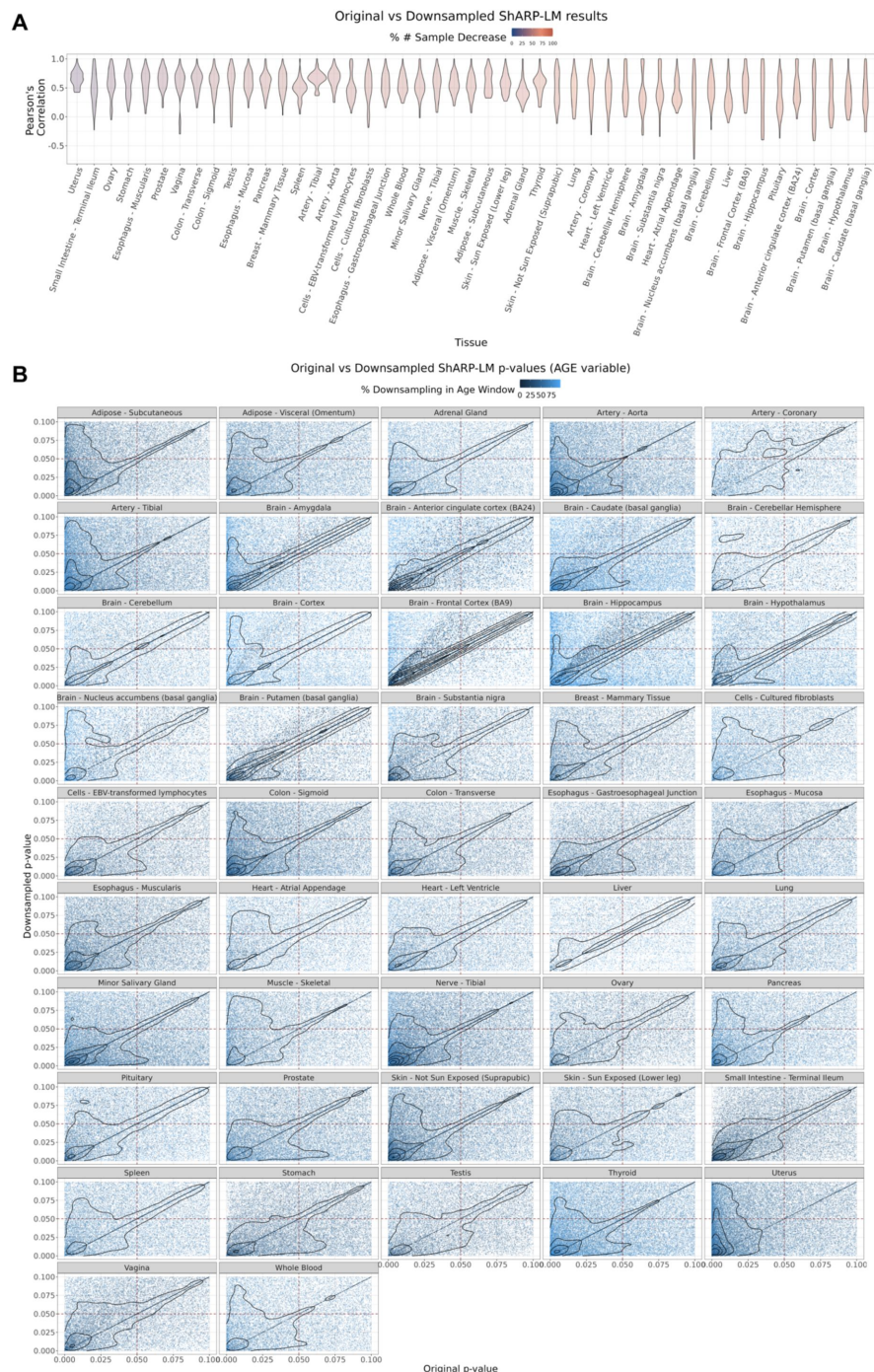
Representation of the expression of Y-chromosome-specific gene *DDX3Y*, stratified by sex (**A**), along with the corresponding significance indicating age-related alterations (**B**). Despite the non-zero expression values for female samples, they fall within a range deemed of background noise (-2 to -8 logCPM). Notably, when assessing the significance of age-related changes in the gene, a distinct peak emerges around the age of 40. This peak is spurious, as it primarily mirrors changes of gene expression in females.



Supplementary Figure 7

Distributions of GTEx donors' ages and its impact on the statistical power of detection of differential gene expression.

For thyroid as an example tissue, the sex-differentiated distribution of GTEx samples' ages (v8) (above) are shown together with the percentage of altered genes over age (below) for the three variables (*Age*, *Sex* and *Age&Sex*), as well as their statistical significance, given by the ShARP-LM approach. The statistical power for detecting transcriptional changes is influenced by the age distribution of donors, leading to a higher likelihood of identifying significant alterations ($FDR \leq 0.05$) in age ranges that are more prevalent in the sample population, typically associated with older individuals.

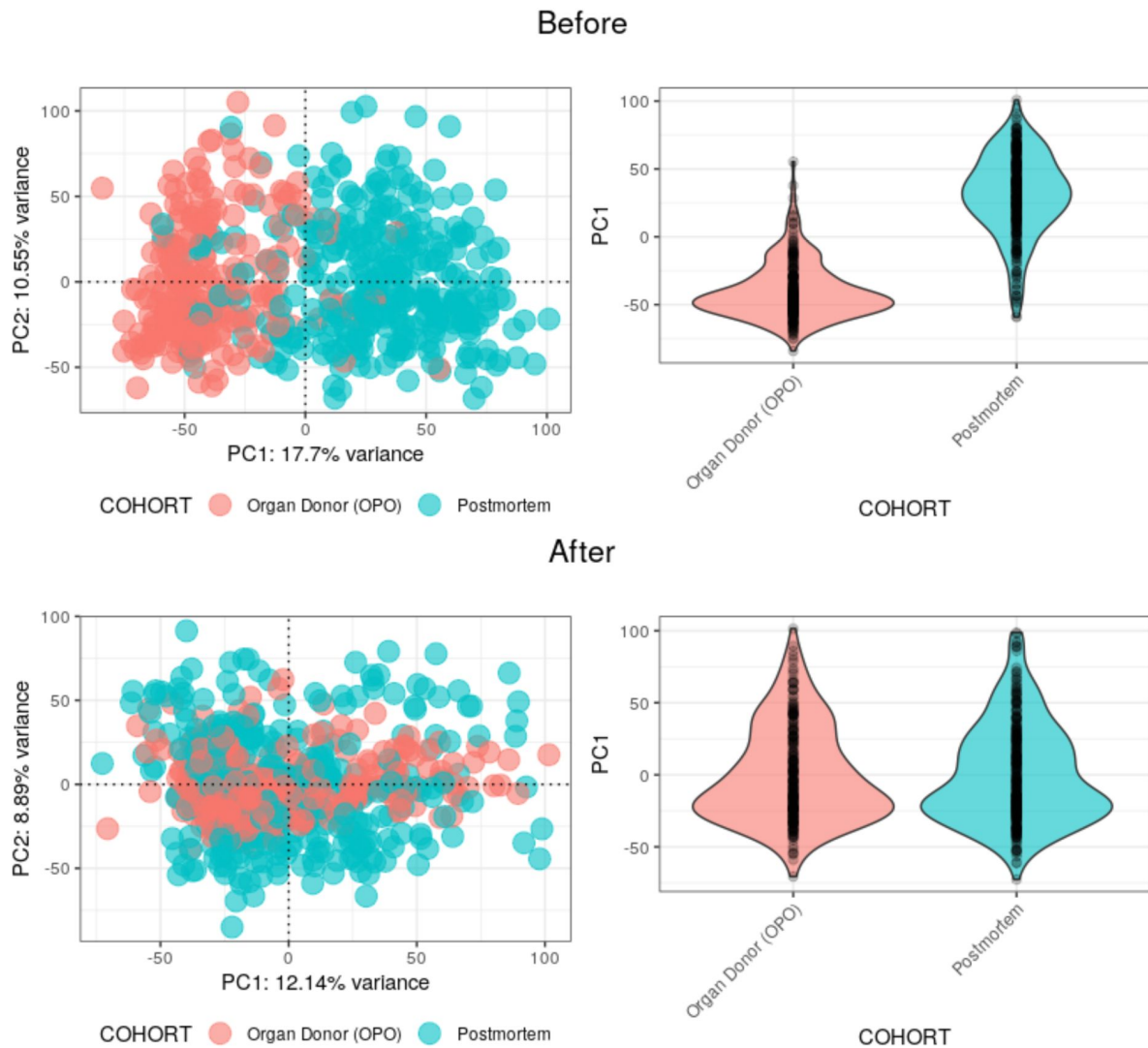


Supplementary Figure 8

Effect of downsampling in ShARP-LM results.

Downsampling was performed by reducing the number of samples in a given window to the minimum available sample size within all age windows for a given tissue. **A** – Per tissue violin plots of gene-wide distributions of Pearson's correlation coefficients between original and downsampled logFC values for the Age variable across age windows, with tissues coloured by and ordered by increasing percentage of downsampling-associated reduction in the number of samples. **B** – Density scatter plots of comparison of associated original and downsampled p-values for each tissue, coloured by the downsampling percentage in each age window, highlighting the low range of p-values (from 0 to 0.1). Despite changes in logFC with downsampling, a considerable correlation in significance is maintained, although downsampling naturally results in a loss of statistical power, evident by the shift of points towards the first quadrant (dashed lines: p-value = 0.05).

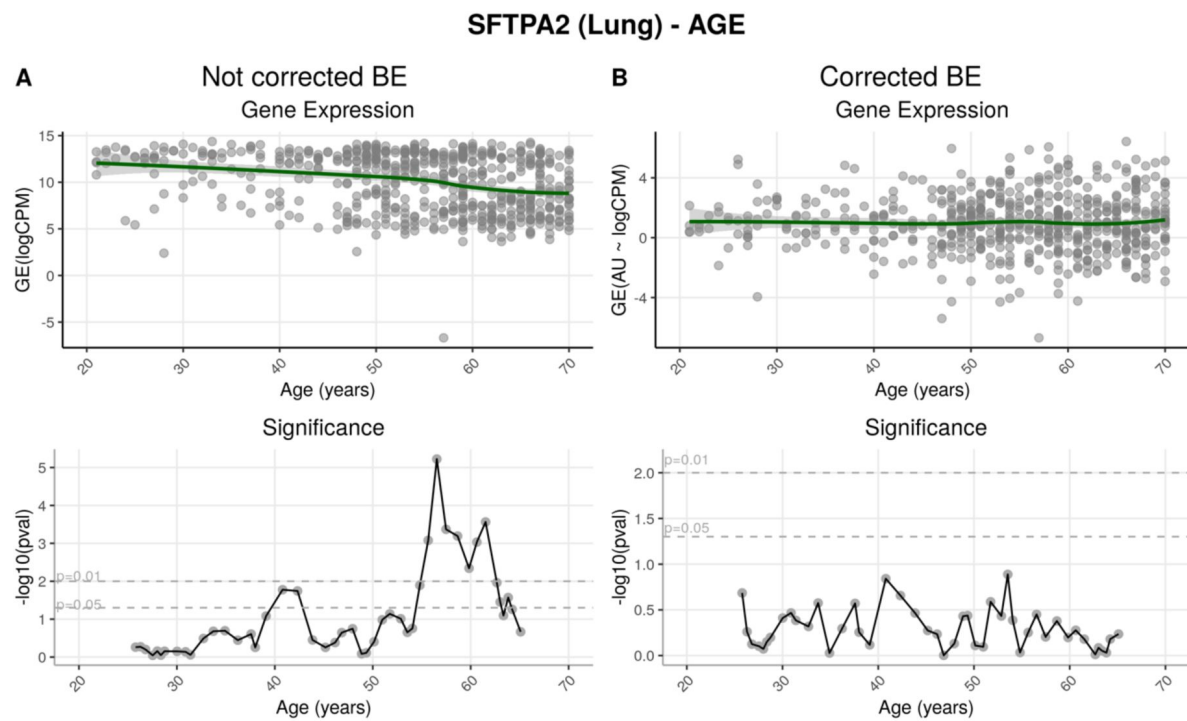
Lung - Batch Effect Correction



Supplementary Figure 9

Batch effect correction applied to lung samples.

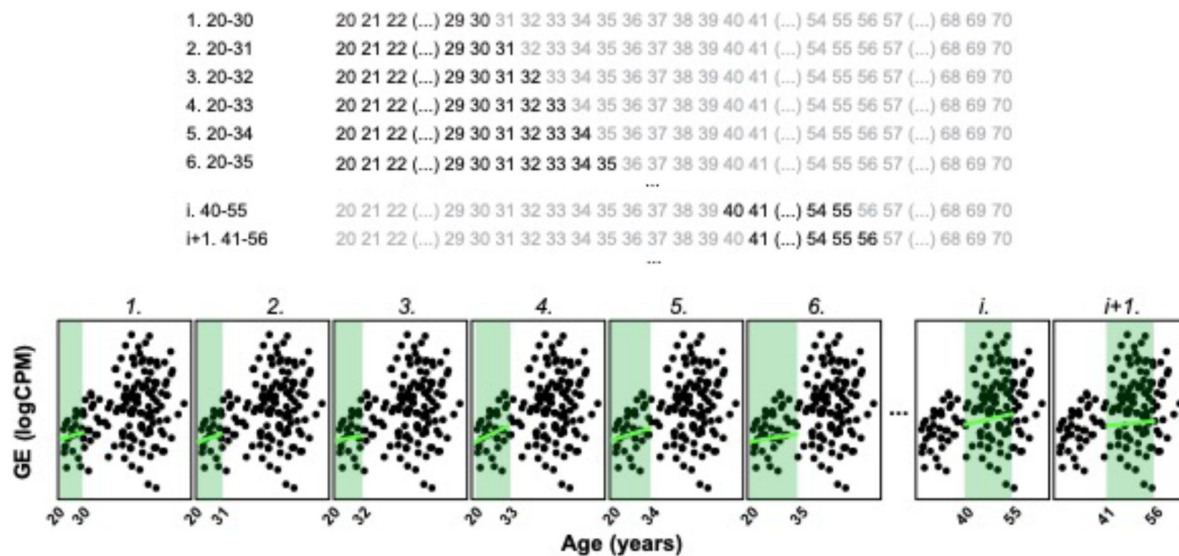
(Top) Before batch effect correction, 17.7% of the variance between samples appears to be mainly explained by the COHORT variable. **(Bottom)** After batch effect correction, PC1 no longer appears to be explained by the COHORT variable, with its weights having a similar distribution between batches.



Supplementary Figure 10

Effects of batch effect correction on gene-centric (*SFTPA2*) analysis.

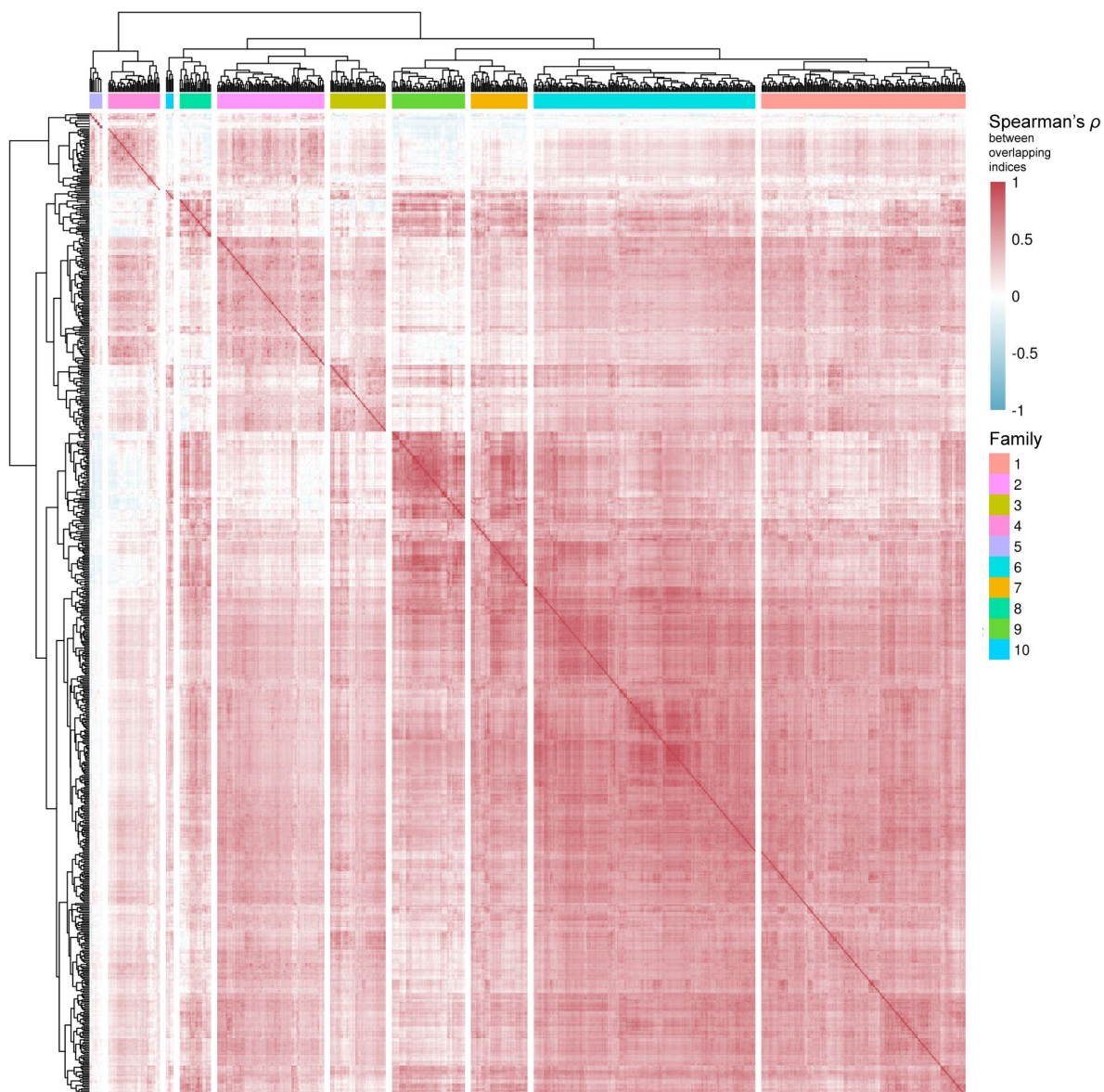
A – Before batch effect correction, *SFTPA2* expression diminishes throughout ageing. This change is particularly significant after around 55 years of age. **B** – After correction, expression of this gene shows no significant alterations across age ranges.



Supplementary Figure 11

Principle of the ShARP-LM method.

ShARP-LM involves fitting linear models to gene expression data for each tissue, with samples from donors grouped into age windows of 16 years each, shifted consecutively through individual years of age using a sliding window approach. As samples at the ends of the dataset's age range would thereby be involved in fewer linear models, we made the window size gradually increase from 11 to 16 years when starting from the "youngest" samples and decrease from 16 to 11 years when reaching the "oldest". The example shows the expression of an arbitrary gene over age, with 8 age windows shown. Coloured rectangles represent the age windows and coloured lines the linear modelling of the gene expression over age in each of them.



Supplementary Figure 12

Clustering of pathways from Reactome and KEGG, and level 3 Gene Ontology Biological Processes.

Based on their common genes (v. Materials and Methods), pathways from curated databases were clustered in 10 families.

Tissue	#Genes
Adipose - Subcutaneous	15519
Adipose - Visceral (Omentum)	15623
Adrenal Gland	15332
Artery - Aorta	15172
Artery - Coronary	15517
Artery - Tibial	14714
Brain - Amygdala	15816
Brain - Anterior cingulate cortex (BA24)	15867
Brain - Caudate (basal ganglia)	16136
Brain - Cerebellar Hemisphere	16207
Brain - Cerebellum	16473
Brain - Cortex	16178
Brain - Frontal Cortex (BA9)	16020
Brain - Hippocampus	15910
Brain - Hypothalamus	16412
Brain - Nucleus accumbens (basal ganglia)	16153
Brain - Putamen (basal ganglia)	15768
Brain - Spinal cord (cervical c-1)	15838
Brain - Substantia nigra	15796
Breast - Mammary Tissue	16311
Cells - Cultured fibroblasts	13737
Cells - EBV-transformed lymphocytes	13743
Colon - Sigmoid	15635
Colon - Transverse	16491
Esophagus - Gastroesophageal Junction	15459
Esophagus - Mucosa	15509
Esophagus - Muscularis	15355
Heart - Atrial Appendage	15003
Heart - Left Ventricle	14036
Kidney - Cortex	16054
Liver	14376
Lung	16616
Minor Salivary Gland	16412
Muscle - Skeletal	13331
Nerve - Tibial	16468
Ovary	15987
Pancreas	14857
Pituitary	17445
Prostate	17065
Skin - Not Sun Exposed (Suprapubic)	16079
Skin - Sun Exposed (Lower leg)	16097
Small Intestine - Terminal Ileum	16901
Spleen	16597
Stomach	15990
Testis	21913
Thyroid	16704
Uterus	15946
Vagina	16538
Whole Blood	13066

Supplementary Table 1

Number of studied genes per tissue.

For each tissue, we kept the genes with at least 1 CPM of expression in at least 40% of the samples.

References

1. Campisi J. (2013) **Aging, Cellular Senescence, and Cancer** *Annu. Rev. Physiol* **75**:685–705
2. Niccoli T., Partridge L (2012) **Ageing as a Risk Factor for Disease** *Curr. Biol* **22**:R741–R752
3. Wyss-Coray T. (2016) **Ageing, neurodegeneration and brain rejuvenation** *Nature* **539**:180–186
4. López-Otín C., Blasco M. A., Partridge L., Serrano M., Kroemer G (2013) **The Hallmarks of Aging** *Cell* **153**:1194–1217
5. Stegeman R., Weake V. M (2017) **Transcriptional Signatures of Aging** *J. Mol. Biol* **429**:2427–2437
6. Tower J. (2017) **Sex-Specific Gene Expression and Life Span Regulation** *Trends Endocrinol. Metab* **28**:735–747
7. Austad S. N., Fischer K. E (2016) **Sex Differences in Lifespan** *Cell Metab* **23**:1022–1033
8. Mele M., et al. (2015) **The human transcriptome across tissues and individuals** *Science* **348**:660–665
9. Gershoni M., Pietrokovski S (2017) **The landscape of sex-differential transcriptome and its consequent selection in human adults** *BMC Biol* **15**
10. Kassam I., Wu Y., Yang J., Visscher P. M., McRae A. F (2019) **Tissue-specific sex differences in human gene expression** *Hum. Mol. Genet* **28**:2976–2986
11. Mayne B. T., et al. (2016) **Large Scale Gene Expression Meta-Analysis Reveals Tissue-Specific, Sex-Biased Gene Expression in Humans** *Front. Genet* **7**
12. Zahn J. M., et al. (2007) **AGEMAP: A Gene Expression Database for Aging in Mice** *PLoS Genet* **3**
13. Kimmel J. C., et al. (2019) **Murine single-cell RNA-seq reveals cell-identity- and tissue-specific trajectories of aging** *Genome Res* **29**:2088–2103
14. Benayoun B. A., et al. (2019) **Remodeling of epigenome and transcriptome landscapes with aging in mice reveals widespread induction of inflammatory responses** *Genome Res* **29**:697–709
15. Lui J. C., Chen W., Barnes K. M., Baron J (2010) **Changes in gene expression associated with aging commonly originate during juvenile growth** *Mech. Ageing Dev* **131**:641–649
16. Shavlakadze T., et al. (2019) **Age-Related Gene Expression Signature in Rats Demonstrate Early, Late, and Linear Transcriptional Changes from Multiple Tissues** *Cell Rep* **28**:3263–3273
17. The Tabula Muris Consortium., 583, (2020) **A single-cell transcriptomic atlas characterizes ageing tissues in the mouse** *Nature* :590–595

18. Schaum N., et al., 583, (2020) **Ageing hallmarks exhibit organ-specific temporal signatures** *Nature* :596–602
19. Zahn J. M., et al. (2005) **Transcriptional profiling of aging in human muscle reveals a common aging signature** *PLoS Genet.*
20. Welle S., Brooks A. I., Delehanty J. M., Needler N., Thornton C. A (2003) **Gene expression profile of aging in human muscle** *Physiol. Genomics* **14**:149–159
21. Jozsi A. C., et al. (2000) **Aged human muscle demonstrates an altered gene expression profile consistent with an impaired response to exercise** *Mech. Ageing Dev* **120**:45–56
22. Rodwell G. E. J., et al. (2004) **A Transcriptional Profile of Aging in the Human Kidney** *PLoS Biol* **2**
23. Galatro T. F., et al. (2017) **Transcriptomic analysis of purified human cortical microglia reveals age-associated changes** *Nat. Neurosci* **20**:1162–1171
24. Olah M., et al. (2018) **A transcriptomic atlas of aged human microglia** *Nat. Commun* **9**
25. Berchtold N. C., et al. (2008) **Gene expression changes in the course of normal brain aging are sexually dimorphic** *Proc. Natl. Acad. Sci* **105**:15605–15610
26. Lu T., et al. (2004) **Gene regulation and DNA damage in the ageing human brain** *Nature* **429**:883–891
27. Işıldak U., Somel M., Thornton J. M., Dönertaş H. M (2020) **Temporal changes in the gene expression heterogeneity during brain development and aging** *Sci. Rep* **10**
28. Haustead D. J., et al. (2016) **Transcriptome analysis of human ageing in male skin shows mid-life period of variability and central role of NF-κB** *Sci. Rep* **6**
29. Holzscheck N., et al. (2020) **Multi-omics network analysis reveals distinct stages in the human aging progression in epidermal tissue** *Aging* **12**:12393–12409
30. Nakamura S., et al. (2012) **Identification of blood biomarkers of aging by transcript profiling of whole blood** *Biochem. Biophys. Res. Commun* **418**:313–318
31. Harries L. W., et al. (2011) **Human aging is characterized by focused changes in gene expression and deregulation of alternative splicing** *Aging Cell* **10**:868–878
32. Thomas R (2002) **Age-Associated Changes in Gene Expression Patterns in the Liver** *J. Gastrointest. Surg* **6**:445–454
33. Yoshida S., Yashar B. M., Hiriyanna S., Swaroop A (2002) **Microarray analysis of gene expression in the aging human retina** *Invest. Ophthalmol. Vis. Sci* **43**:2554–60
34. Gheorghe M., et al. (2014) **Major aging-associated RNA expressions change at two distinct age-positions** *BMC Genomics* **15**
35. Yang J., et al. (2015) **Synchronized age-related gene expression changes across multiple tissues in human and the link to complex diseases** *Sci. Rep* **5**

36. Aramillo Irizar P., et al. (2018) **Transcriptomic alterations during ageing reflect the shift from cancer to degenerative diseases in the elderly** *Nat. Commun* **9**
37. Glass D., et al. (2013) **Gene expression changes with age in skin, adipose tissue, blood and brain** *Genome Biol* **14**
38. Yu Y., et al. (2014) **A rat RNA-Seq transcriptomic BodyMap across 11 organs and 4 developmental stages** *Nat. Commun* **5**
39. Rando T. A., Wyss-Coray T (2021) **Asynchronous, contagious and digital aging** *Nat. Aging* **1**:29–35
40. Lonsdale J., et al. (2013) **The Genotype-Tissue Expression (GTEx) project** *Nat. Genet* **45**:580–585
41. Van Deursen J. M (2014) **The role of senescent cells in ageing** *Nature* **509**:439–446
42. Gorgoulis V., et al. (2019) **Cellular Senescence: Defining a Path Forward** *Cell* **179**:813–827
43. Gil J., Peters G (2006) **Regulation of the INK4b–ARF–INK4a tumour suppressor locus: all for one or one for all** *Nat. Rev. Mol. Cell Biol* **7**:667–677
44. Erickson S., et al. (1998) **Involvement of the Ink4 proteins p16 and p15 in T-lymphocyte senescence** *Oncogene* **17**:595–602
45. Narita M., et al. (2003) **Rb-Mediated Heterochromatin Formation and Silencing of E2F Target Genes during Cellular Senescence** *Cell* **113**:703–716
46. Sun X., Kaufman P. D (2018) **Ki-67: more than a proliferation marker** *Chromosoma* **127**:175–186
47. Kaczmarek M (2014) **Variation in Age at Natural Menopause among Polish Women in Relation to Biological and Social Factors** *Problems in Past and Present Populations, Biennial Book of EAA* <https://doi.org/10.13140/2.1.2610.1449>
48. Croft D., et al. (2014) **The Reactome pathway knowledgebase** *Nucleic Acids Res* **42**:D472–D477
49. Subramanian A., et al. (2005) **Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles** *Proc. Natl. Acad. Sci. U. S. A* **102**:15545–50
50. Kanehisa M (2000) **KEGG: Kyoto Encyclopedia of Genes and Genomes** *Nucleic Acids Res* **28**:27–30
51. Gene Ontology Consortium (2004) **The Gene Ontology (GO) database and informatics resource** *Nucleic Acids Res* **32**:258–261
52. Langfelder P., Horvath S (2008) **WGCNA: an R package for weighted correlation network analysis** *BMC Bioinformatics* **9**
53. van Dam S., Vösa U., van der Graaf A., Franke L., de Magalhães J. P (2017) **Gene co-expression analysis for functional classification and gene-disease predictions** *Brief. Bioinform.* <https://doi.org/10.1093/bib/bbw139>

54. Kelley KW, Nakao-Inoue H, Molofsky AV O. M (2018) **Variation among intact tissue samples reveals the core transcriptional features of human CNS cell classes** *Nat. Neurosci* **21**:1171–1184
55. Piñero J., et al. (2019) **The DisGeNET knowledge platform for disease genomics: 2019 update** *Nucleic Acids Res.* <https://doi.org/10.1093/nar/gkz1021>
56. Piñero J., et al. (2017) **DisGeNET: a comprehensive platform integrating information on human disease-associated genes and variants** *Nucleic Acids Res* **45**:D833–D839
57. Palmer A. L., Ousman S. S (2018) **Astrocytes and Aging** *Front. Aging Neurosci* **10**
58. Lehallier B., et al. (2019) **Undulating changes in human plasma proteome profiles across the lifespan** *Nat. Med* **25**:1843–1850
59. Khramtsova E. A., Davis L. K., Stranger B. E (2019) **The role of sex in the genomics of human complex traits** *Nat. Rev. Genet* **20**:173–190
60. Subramanian A., et al. (2017) **A Next Generation Connectivity Map: L1000 Platform and the First 1,000,000 Profiles** *Cell* **171**:1437–1452
61. de Almeida BP, Saraiva-Agostinho N (2020) **cTRAP: Identification of candidate causal perturbations from differential gene expression data**
62. de Magalhães J. P., Curado J., Church G. M (2009) **Meta-analysis of age-related gene expression profiles identifies common signatures of aging** *Bioinformatics* **25**:875–881
63. Tacutu R., et al. (2018) **Human Ageing Genomic Resources: new and updated databases** *Nucleic Acids Res* **46**:D1083–D1090
64. Craig T., et al. (2015) **The Digital Ageing Atlas: integrating the diversity of age-related changes into a unified resource** *Nucleic Acids Res* **43**:D873–D878
65. Dönertaş H. M., Fuentealba Valenzuela M., Partridge L., Thornton J. M (2018) **Gene expression-based drug repurposing to target aging** *Aging Cell* **17**
66. Janssens G. E., et al. (2019) **Transcriptomics-Based Screening Identifies Pharmacological Inhibition of Hsp90 as a Means to Defer Aging** *Cell Rep* **27**:467–480
67. Chang W, Cheng J, Allaire J, Sievert C, Schloerke B, Xie Y, Allen J, McPherson J, Dipert A, Borges B **Chang W, Cheng J, Allaire J, Sievert C, Schloerke B, Xie Y, Allen J, McPherson J, Dipert A, Borges B shiny: Web Application Framework for R** <https://shiny.rstudio.com/gallery/>.
68. Kunst J. **highcharter: A Wrapper for the ‘Highcharts’ Library**
69. Yihui X., Joe C., Xianying T. **DT: A Wrapper of the JavaScript Library ‘DataTables’**
70. Robinson M. D., McCarthy D. J., Smyth G. K (2010) **edgeR: a Bioconductor package for differential expression analysis of digital gene expression data** *Bioinformatics* **26**:139–140
71. Robinson M. D., Oshlack A (2010) **A scaling normalization method for differential expression analysis of RNA-seq data** *Genome Biol* **11**

72. Law C. W., Chen Y., Shi W., Smyth G (2014) **K. voom: precision weights unlock linear model analysis tools for RNA-seq read counts** *Genome Biol* **15**
73. Ritchie M. E., et al. (2015) **limma powers differential expression analyses for RNA-sequencing and microarray studies** *Nucleic Acids Res* **43**
74. García-Pérez R., et al. (2023) **The landscape of expression and alternative splicing variation across human traits** *Cell Genomics* **3**
75. Buuren S. V., Groothuis-Oudshoorn K. (2011) **mice: Multivariate Imputation by Chained Equations in R** *J. Stat. Softw* **45**
76. McCall M. N., Illei P. B., Halushka M. K (2016) **Complex Sources of Variation in Tissue Expression Data: Analysis of the GTEx Lung Transcriptome** *Am. J. Hum. Genet* **99**:624–635
77. Korotkevich G., et al. (2021) **Fast gene set enrichment analysis** *bioRxiv* <https://doi.org/10.1101/060012>
78. GSEA-MSIGDB **GSEA-MSIGDB. Gene Set Enrichment Analysis (GSEA) User Guide.**
79. Chen Y.-A., et al. (2014) **Integrated Pathway Clusters with Coherent Biological Themes for Target Prioritisation** *PLoS ONE* **9**
80. Dowle M., Srinivasan A. **data.table: Extension of ‘data.frame’**
81. Langfelder P., Horvath S. **Langfelder, P. & Horvath, S. WGCNA Frequently Asked Questions.**
82. Chen B., Khodadoust M. S., Liu C. L., Newman A. M., Alizadeh A. A. (2018) **Profiling Tumor Infiltrating Immune Cells with CIBERSORT** *Cancer Systems Biology* **1711**:243–259
83. Regev A., et al. (2017) **The Human Cell Atlas** *eLife* **6**
84. Fan X., et al. (2018) **Spatial transcriptomic survey of human embryonic cerebral cortex by single-cell RNA-seq analysis** *Cell Res* **28**:730–745
85. Cao J., et al. (2020) **A human cell atlas of fetal gene expression** *Science* **370**

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Reviewer #1 (Public Review):

This fascinating paper by A.L. Schneider et al. describes voyAGER, a shiny-based interface for easy exploration of the GTEx dataset by non- or novice programmers. Importantly, voyAGER is open source and available from github, which could greatly accelerate additional development and further uses of this interesting tool.

The authors developed a pipeline for modeling age-related changes in gene expression in the GTEx data called ShARP-LM, fitting a linear model for age, sex and age&sex interaction terms. This pipeline underlies the later analyses that can be applied within voyAGER. These analyses are labeled by tissue so that users can easily begin a query based on a tissue or a gene of possible interest.

voyAGER implements many kinds of interesting R-based tools such as pathway overrepresentation analysis and gene co-expression module analysis, in a way that akes these approaches accessible to non-bioinformaticist aging researchers.

As the tidal wave of publicly available large, high-dimensional datasets such as transcriptomes continues to grow exponentially, the usefulness of tools such as voyAGER will only increase. While test users may be able to imagine features or refinements they wish were already present, due to the open source approach they or anyone else including but not limited to the present authors can implement additional features in the future. I look forward to using this tool and to staying abreast of its future development.

Overall, this study describes a new tool of interest to the field. The manuscript is clearly written overall, with a few minor suggested corrections, as noted below. The figures and supplementary information are all clear and all add to the manuscript.

<https://doi.org/10.7554/eLife.88623.2.sa2>

Reviewer #2 (Public Review):

The purpose of this study is to develop a tool that serves as a starting point for investigating and uncovering genes and pathways associated with aging. The tool utilizes information from the GTEx public database, which contains post-mortem human data. It focuses on identifying age-related gene expression changes across different age range, biological sexes, and medical histories, with a focus on specific tissues.

Additionally, the authors envision the platform as continuously evolving, with ongoing development and expansion to include new data and features, ensuring it remains a cutting-edge resource for researchers studying aging.

voyAGER presents a tool for exploring gene expression changes across multiple tissues in the context of aging. One of the main strengths of the tool is its intuitive and user-friendly interface, which allows for easy navigation and exploration of gene expression patterns for biologists. Users can explore changes in gene expression of single genes across multiple tissues, enabling them to identify genes of interest that can be further investigated.

A particularly noteworthy strength of the tool is its ability to show tissue-specific gene expression patterns. This feature is essential for elucidating the paradigm of tissue-specific asynchronous aging and provides a unique and valuable resource for the aging community.

However, the choice of the R shiny platform for visualization may not be the most conducive to extensibility and open-source collaboration, owing to its lack of modularity. Alternatives like Flask or FastAPI, which are more production-oriented, could be more appropriate. Additionally, despite using preprocessed data and functioning primarily as a visualization platform, the tool occasionally experiences lag, indicating room for performance improvement. These aspects are worth considering for future versions of the tool.

Overall, voyAGER offers an entry point for further investigation of genes involved in aging, and its ability to show tissue-specific gene expression patterns provides a unique and valuable resource for the scientific community.

Finally, the tool is complemented by a comprehensive tutorial that elucidates each functionality and includes examples. The authors have shared the code for preprocessing and the tool itself. They also acknowledge the limitations of the statistical inference tests and their interpretation in the manuscript, contributing to its transparency.

<https://doi.org/10.7554/eLife.88623.2.sa1>

Reviewer #3 (Public Review):

In their manuscript, Schneider et al. aim to develop voyAGER, a web-based tool that enables the exploration of gene expression changes over age in a tissue- and sex-specific manner. The authors achieved this goal by calculating the significance of gene expression alterations within a sliding window, using their unique algorithm, Shifting Age Range Pipeline for Linear Modelling (ShARP-LM), as well as tissue-level summaries that calculated the significance of the proportion of differentially expressed genes by the windows and calculated enrichments

of pathways for showing biological relevance. Furthermore, the authors examined the enrichment of cell types, pathways, and diseases by defining the co-expressed gene modules in four selected tissues. Although their algorithm ShARP-LM has limited statistical power due to its calculation within a 16-year window, the voyAGER was developed as a discovery tool, giving researchers easy access to the vast amount of transcriptome data from the GTEx project. Overall, the research design is unique and well-performed in simulating age-dependent changes in gene expression. The interesting results provide useful resources for the field of human genetics of aging.

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Author Response

The following is the authors' response to the original reviews.

We are pleased to send you a revised version of our manuscript entitled “voyAGER: free web interface for the analysis of age-related gene expression alterations in human tissues” and the associated shiny web app, in which we incorporate the referees' feedback. We would like to express our gratitude for their time and valuable insights, which have contributed to the improvement of our work. We appreciate the rigorous evaluation process that eLife maintains.

In this letter, we address each of the reviewers' comments and concerns, point-by-point, offering detailed responses and clarifications. We have made several revisions to our manuscript following their recommendations.

We must note that the revised version of the manuscript has two novel joint first authors, Rita Martins-Silva and Alexandre Kaizeler, who performed all the requested reanalyses, given that the initial first author, Arthur Schneider, already left our lab. We must also point to the following minor unsolicited improvements we took the opportunity to make:

- Added a comprehensive tutorial to the GitHub repository on how to navigate through voyAGER's features.
- Implemented sample randomisation in the scatter plots depicting gene expression across the age axis to ensure data privacy.
- Implemented minor adjustments within the web app to enhance user comprehension and clarity when visualizing the data.
- Improved clarity of the methodological sections.

Reviewer 1

(1.1) While this may be obvious to others for some reason that escaped me, I was unsure what was the basis for the authors' choice of 16 years as the very specific sliding window size. If I'm not alone in this, it might add clarity for other readers and users if this parameter choice were explained and justified more explicitly.

We apologise for our omission in providing the rationale behind our choice in the previous version. We chose 16 years as our sliding window size because this was the minimum needed to guarantee the presence of more than one sample per window, across all the tissues considered in the study (Figure R1 below).

We added the following sentence to the manuscript (v. Methods, ShARP-LM):

“This was the minimum age span needed to guarantee the presence of more than one sample per window, across all considered tissues.”

(1.2) *"In particular, tissue-specific periods of major transcriptional changes in the fifth and eighth decades of human lifespan have been revealed, reflecting the so-called digital aging and consistently with what is observed in mice" here I think that "consistently" should be "consistent".*

We thank the reviewer for the comment and following the suggestion, we have revised 'Consistently' to 'consistent' as it is the correct usage in our sentence.

(1.3) *"On a different note, sex biases have been reported in for the expression of SALL1 and KAL1 in adipose tissue and lung, respectively." Here I think that "in for" should be "in".*

As recommended by the reviewer, we have replaced 'in for' for 'in'. As we substituted KAL1, the current sentence now stands as “On a different note, sex biases have been reported in the expression of SALL1 and DDX43 in adipose tissue and lung, respectively”.

(1.4) *"We downloaded the matrix with the RNA-seq read counts for each gene in each GTEx v7 sample from the project's data portal (<https://www.gtexportal.org/>)."* In my pdf manuscript this hyperlink appears to be broken.

We appreciate the reviewer's attention to the broken link, and we have rectified the issue. The link should now be fully operational, effectively directing users to the GTEx Portal.

(1.5) *Under methods, I might suggest "Development platform" or "Development platforms" over "Development's platform" as a heading.*

We have modified the heading of this section in the methods to 'Development Platforms', as we believe it better reflects the information conveyed.

Reviewer 2

(2.1) *In this tool/resource paper, it is crucial that the data used is up-to-date to provide the most comprehensive and relevant information to users. However, the authors utilized GTEx v7, which is an outdated (2016) version of the dataset. It is worth noting that GTEx v8 includes over 940 individuals, representing a 35% increase in individuals, and a 50% increase in the total number of samples. The authors should check the newer versions of GTEx and update the data.*

When the development of the voyAGER web application began, GTEx version 7 was the most up to date. Nevertheless, we agree that the version 8 offers a notably more extensive dataset, encompassing a larger number of individuals, samples, and introducing new tissues. Consequently, we have updated our application to incorporate the data from GTEx version 8.

(2.2) *The authors did not address any correction for batch effects or RNA integrity numbers, which are known to affect transcriptome profiles. For instance, our analysis of GTEx v8 Cortex tissue revealed that after filtering out lowly expressed genes, in the same way authors did, PC1 (which accounts for 24% of the variation) had a Spearman's correlation value of 0.48 ($p < 6.1e-16$) with RNA integrity number.*

We acknowledge the validity of the reviewer's comment and appreciate the importance of such corrections to enhancing data interpretation. In response, we conducted a thorough

unbiased investigation into potential batch effects, with the COHORT variable emerging as the primary driver of those observed across most tissues. Furthermore, SMRIN (as the reviewer pointed), DTHHRDY, MHSMKYRS and the number of detected genes in each sample were consistently associated with the primary sources of variation. As a result, we implemented batch effect correction for those five conditions, in a tissue-specific manner.

We provide a detailed explanation of the batch effect correction methodology and its importance in the biological interpretation of results in the Methods section, specifically under "Read count data pre-processing". Additionally, we have included two new supplementary figures, Sup. Figures 7 and 8, to illustrate a batch effect example in lung tissue and emphasise the critical role of this correction in data interpretation.

(2.3) The data analyzed in the GTEx dataset is not filtered or corrected for the cause of death, which can range from violent and sudden deaths to slow deaths or cases requiring a ventilator. As a result, the data may not accurately represent healthy aging profiles but rather reflect changes in the transcriptome specific to certain diseases due to the age-related increase in disease risk. While the authors do acknowledge this limitation in the discussion, stating that it is not a healthy cohort and disease-specific analysis is not feasible due to the limited number of samples, it would be useful for users to have the option to analyze only cases of fast death, excluding ventilator cases and deaths due to disease. This is typically how GTEx data is utilized in aging studies. Alternatively, the authors should consider including the "cause of death" variable in the model.

This comment is closely related to the prior discussion (point 2.2). Notably, two of the covariates selected for batch effect correction, namely, DTHHRDY (Death classification based on the 4-point Hardy Scale¹) and COHORT (indicating whether the participant was a postmortem, organ, or surgical donor¹), have a direct relevance to this issue, i.e., both relate to the cause of death of the individual.

¹ According to the nomenclature of variables described in https://www.ncbi.nlm.nih.gov/projects/gap/cgi-bin/GetListOfAllObjects.cgi?study_id=phs000424.v9.p2&object_type=variable

We therefore effectively account for their influence on gene expression, mitigating these factors' impact.

This approach represents a compromise, as it is practically infeasible to ascertain the absence of underlying health conditions in the remaining samples, even if only considering cases of "fast death". Hence, we opted to keep all samples, independently of the cause of death of its donor, to dilute potential effects associated with individual causes of death.

(2.4) The age distribution varies across tissues which may impact the results of the study. The authors' claim that age distribution does not affect the outcomes is inconclusive. Since the study aims to provide cross-tissue analysis, it is important to note that differing age distributions across tissues can influence the overall results. To address this, the authors should conduct downsampling to different age distributions across tissues and evaluate the level of tissue-specific or common changes that remain after the distributions are made similar.

We acknowledge that variations in age distributions are evident across different tissues, with brain tissues displaying a notably pronounced disparity (green density lines in Figure R2 below).

To address this issue comprehensively, we conducted tissue-specific downsampling, by reducing the number of samples in a given age window to the minimum available sample size within all age windows for a given tissue. The histograms (density plots) of the number of samples per age window of 16 years considered in the ShARP-LM model, as well as the

minimum number of samples in each age window, per tissue are illustrated in Figure R1. After performing downsampling, we computed the logFC and p-value of differential expression for each gene, per age window, and compared them (for all genes in a given age window) with those involving all samples.

Despite changes in logFC with downsampling, a considerable positive correlation is maintained (Figure R3, top panel). This suggests that the overall trends in gene expression changes persist. However, the downsampling process expectedly results in a decrease of statistical power within each age window concomitant with the decreased sample size, evident from the shift of genes from the third to the first quadrant in Figure R3, bottom panel. Consequently, we have opted for maintaining results encompassing all samples and removing the paragraph in the Discussion that asserted the absence of age distribution impact on the overall outcomes (“Indeed, we found no confounding between the distribution of samples’ ages and the trend of gene expression progression over age in any tissue.”), as we deem it inaccurate, potentially leading to misinterpretation. We have added a supplementary figure (Supplementary Figure 8, identical to Figure R3) illustrating the effect of downsampling, and the following paragraph to the manuscript’s Discussion section:

“When downsampling to ensure a balanced age distribution, a loss of statistical power is apparent but a considerable positive correlation with the original results is maintained and a substantial number of significant alterations remain so (Supplementary Figure 8).”

We acknowledge that this limitation can be addressed with the growing accumulation of human tissue transcriptomes in publicly available databases, a trend we anticipate in the near future. We are committed to promptly updating voyAGER with any new data releases that may offer a solution to this concern.

Nonetheless, we want to underscore, as the reviewer has astutely pointed out, that while voyAGER can facilitate cross-tissue comparisons, it must be done with caution. In this regard, we inserted the following paragraph into the Discussion:

“Due to the tissue-specific nature of the pre-processing steps (v. Read count data preprocessing in the Methods section), and given that most of the plotted gene expression distributions are centred and scaled by tissue, it is important to note that voyAGER may not be always suited for direct comparisons between different tissues. For instance, it does not allow to directly ascertain if a gene exhibits different expression levels in different tissues or if the expression of a particular gene in one tissue changes more drastically with age than in another tissue.”

(2.5) The GTEx resource is extremely valuable, however, it comes with challenges. GTEx contains tissue samples from the same individuals across different tissues, resulting in varying degrees of overlap in sample origin across tissues as not all tissues are collected for all individuals. This could affect the similar/different patterns observed across tissues. As this tool is meant for broader use by the community, it is crucial for the authors to either rule out this possibility by conducting a cross-tissue comparison using a non-parametric model that accounts for the dependency between samples from the same individual, or to provide information on the degree of similarity between samples so that the users can keep this possibility in mind when using the tool for hypothesis generation.

We agree that the variable degrees of overlap between tissues (Figure R4) could lead to a confounding between trends in a population of common individuals and those associated with age. We therefore examined the contributions of variables ‘donor,’ ‘tissue,’ and ‘age’ to the overall variance in the data (Figure R5, panel A), having normalised the data collectively across all tissues. Tissue and donor contribute approximately 90% and 10% of the variance, respectively. Age exhibits minimal impact (around 1%), which may be attributed to the relative subtlety of its effects on gene expression and to the tissue specificity of ageing-

associated changes. Notably, removing the 'donor' variable does not transfer this variance to 'age', suggesting a limited confounding between these variables (see Figure R5, panel B).

We also specifically examined the pairs of tissues exhibiting the lowest (Brain Amygdala / Small Intestine), median (Pancreas / Heart Left Ventricle), and highest (Kidney Cortex / Muscle Skeletal) percentages of shared donors. We identified and selectively removed samples from shared donors while maintaining the original sample size imbalance between tissues. Subsequently, we calculated each gene's mean expression within each age window from the ShARP-LM pipeline, followed by each gene's Pearson's correlation of expression between tissue pairs. The resulting coefficients, both with and without the removal of common donors, were compared in scatter plots (Figure R6, left plots). As this process inherently involves downsampling, which may impact results (v. comment 2.4), we performed additional downsampling by randomly removing samples from both tissues according to the proportions defined for the removal of common donors (Figure R6, right plots).

In the chosen scenarios, we note a similar impact between the targeted removal of common donors and random downsampling. Nevertheless, the effects of removing samples may vary according to the absolute number of remaining samples. Consequently, singling out individual cases may not provide conclusive insights. To systematically address this, we represented all tissue pairs in a heatmap, colour-coded based on whether the removal of common donors is more impactful (red) or less impactful (blue) than random downsampling (Figure R7). The values depicted in the heatmap, denoted as the Impact of Common Donors (ICD), are computed for each tissue pair. This calculation involves several steps: first, we determined the absolute difference in Pearson's correlation for each gene's mean expression within each age window from the ShARP-LM pipeline, between the original data and the subset of data without common donors (DiffWoCD) or with random downsampling (DiffRD). Subsequently, the medians of DiffWoCD and DiffRD are computed, and the difference between these median values provides the ICD for each tissue pair. Due to the unidirectional nature of correlation (i.e., the results for tissue 1 vs tissue 2 mirror those for tissue 2 vs tissue 1), the resulting matrix is triangular in form.

We have added a supplementary figure (Supplementary Figure 4, a composition of Figures R4-R7, together with a scatterplot relating the values of heatmaps R4 and R7) that aims to provide guidance to users when interpreting specific tissue pairs, acknowledging inherent limitations (refer to comment 2.4). We have also inserted the following paragraph into the manuscript's Discussion section:

“Furthermore, we must emphasise that the majority of GTEx donors contributed samples to multiple tissues (Supplementary Figure 4A), potentially introducing biases and confounders when comparing gene expression patterns between tissues. Our analyses of variance (Supplementary Figure 4B) and downsampling to control for common donors (Supplementary Figures 4C-E) suggest very limited global confounding between the impacts of donor and age on gene expression and that any potential cross-tissue bias not to depend much on the proportion of common donors (Supplementary Figure 4E). However, this effect must be taken into account when comparing specific pairs of tissues (e.g., Colon – Transverse and Whole Blood, Supplementary Figure 4D).”

(2.6) The authors aimed to create an open-source and ever-evolving resource that could be adapted and improved with new functionality. However, this goal was only partially achieved. Although the code for the web app is open source, crucial components such as the statistical tests or the linear model are not included in the repository, limiting the tool's customizability and adaptability.

We greatly appreciate the reviewer's concern and share their commitment to maintaining the principles of openness, reproducibility, and adaptability for voyAGER. voyAGER was primarily designed as a visualisation tool, displaying pre-processed results, and indeed only the code for the Shiny app itself was accessible through the project's GitHub repository.

To address this shortcoming, we have made the entire data preprocessing script publicly available in the GitHub repository of voyAGER. This script encompasses, among others, filtration, normalisation, batch effect correction, the ShARP-LM pipeline and statistical tests employed, and module definition. Moreover, the web app itself offers functionality to export relevant plots and tables.

(2.7) Furthermore, the authors' choice of visualization platform (R shiny) may not be the best fit for extensibility and open-source collaboration, as it lacks modularity. A more suitable alternative could be production-oriented platforms such as Flask or FastAPI.

We appreciate this thoughtful concern. The decision to use Shiny was primarily driven by our data having already been prepared in the R environment during pre-processing steps. Consequently, and as the web app serves the purpose of visualisation only (and not data processing), Shiny is as a natural and convenient extension of our scripts, enabling data visualisation seamlessly.

We acknowledge that Shiny may lack the modularity required for optimal open-source collaboration. While we recognise the merits of alternative platforms like Flask or FastAPI, we decided to keep Shiny because the current iteration of voyAGER offers significant value to the community. Transitioning to a different platform would be a time-consuming endeavour, that would postpone the release of such resource.

However, the reviewer's feedback regarding modularity and open-source collaboration is duly noted and highly valuable. We will certainly take it into account when developing new web applications within our laboratory.

(2.8) To facilitate collaboration and improve the tool's adaptability, data resulting from the preprocessing pipeline should be made publicly available. This would make it easier for others to contribute and extend the tool's functionality, ultimately enhancing its value for the scientific community.

As outlined in point 2.6 of this rebuttal letter, certain metadata used in our analysis are subject to restricted access. To address this, we have taken several measures to foster transparency and reproducibility of our analyses. First, we have made the scripts for data pre-processing publicly available, along with a comprehensive explanation of our methodology within the main manuscript. This empowers users to replicate our analyses and provides a foundation for those interested in contributing to the tool's development. Furthermore, we have created new issues on voyAGER's GitHub repository, outlining novel features and improvements we envision for the application in the future. We actively encourage users to engage with this section.

(2.9) It is unfortunate that the manuscript has no line numbers, which makes pointing out language issues or typos cumbersome. Below are some minor typos present in the current version mostly due to inconsistent usage of British vs US English, and the authors would be advised to do a thorough proofreading for the final submission.

- *Page 12: Inconsistent spelling of "analyzed" and "analysed". Should be "analyzed", since US English is used throughout the rest of the paper.*

- Page 14: "randomised"
- Page 15: "emphasise"

We apologise for it and include line numbers in the revised version. We have opted for British English and corrected the manuscript accordingly.

(2.10) Some figures in the supplemental material have a low resolution (e.g. S. Fig 5). Especially figures that are not based on screenshots would ideally be of a higher resolution.

As voyAGER is designed as a web application for visualisation, it is inherent that some screenshots of the final resource may have lower resolutions. In response to this concern, we re-generated the figures in this manuscript with a resolution that maintains clarity and readability. We also recreated figures not derived from screenshots, further improving their resolution.

We saved all figures in PDF format and are sending them together with this letter and the revised manuscript, to address any potential issues related to low-resolution figures that may occur during the export of the Word document.

<(2.11) In Fig. 1 in the bottom row the sex labels are hard to see.

We have adapted the figure to address this concern.

(2.12) Math symbols and equations are not well formatted. For example, the GE equation on p. 13, or Oij equation should be properly typeset. Also, the Oij notation might be confusing, I believe the authors meant to use a capital "I", i.e. OI_{ij}.

We have incorporated these recommendations into the revised manuscript.

(2.13) The Readme file in the git repo is very short. It would be helpful to have build and run instructions.

We have updated the README file in the GitHub repository, which now contains, among other features, instructions for launching the Shiny app and building the associated Docker image. Additionally, a simple tutorial has also been included to assist users in navigating through voyAGER's functionalities.

(2.14) "Module" tab's UI inconsistent to other tabs (i.e. "Gene" and "Tissue"), since it contains an "About" page. Adding the "About" page in the actual "Module" page might make the UI clearer.

We believed that the Modules section, due to its distinct methodology, would benefit from an additional tab explaining its underlying rationale. We relate to the reviewer's concern regarding the use of tabs throughout the application and made changes to the app in order to ensure consistency.

(2.15) I would suggest changing the type of the article to "Tools and Resources".

We agree and followed the reviewer's suggestion.

Reviewer 3

(3.1) In the gene-centric analyses section of the result, to improve this manuscript and database, linear regression tests accounting for the entire range of age should be added. The authors' algorithm, ShARP-LM, tests locally within a 16-year window which makes it has lower power than the linear regression test with the whole ages. I suspect that the power reduction is strongly affected in the younger age range since a larger number of GTEx donors are enriched in old age. By adding the results from the lm tests, readers would gain more insight and evidence into how significantly their interest genes change with age.

We are grateful for the reviewer's thoughtful and pertinent recommendation and have thus conducted linear regression tests covering the entire age range. The outcomes of these tests have been integrated into the web application, denoted by a dotted orange line on the 'Gene Expression Alterations Over Age' plots. Additionally, a summary of statistics of overall changes, encompassing p-values, t-statistics, and logFC per year, has been included below the plot title. We have also updated the manuscript to include such changes (v. Methods, Gene-centric visualisation of tissue-specific expression changes across age):

“We also applied a linear model across the entire age range, thereby providing users with more insight and supporting evidence into how a specific gene changes with age. For visualisation purposes, we incorporated a dashed orange line, with the logFC per year for the Age effect as slope, in the respective scatter plots (Figure 3B c). We depict the Sex effect therein by prominent dots on the average samples, with pink and blue denoting females and males, respectively.”

Concerning the observation about the potential reduction in statistical power due to the limited number of samples in younger ages, we acknowledge its validity. Indeed, we have addressed this issue in the manuscript's Discussion (v. Supplementary Figure 6).

(3.1) In line with the ShARP-LM test results, it is not clear which criterion was used to define the significant genes and the following enrichment analyses. I assume that the criterion is $P < 0.05$, but it should be clearly noted. Additionally, the authors should apply adjusted p-values for multiple-test correction. The ideal criterion is an adjusted $P < 0.05$. However, if none or only a handful of genes were found to be significant, the authors could relax the criteria, such as using a regular $P < 0.01$ or 0.05 .

We apologise for any confusion regarding the terminology "significant genes." Our choice to use nonadjusted p-values for determining the significance of gene expression changes with Age, Sex, and their interaction was deliberate, and we would like to clarify our reasoning:

(1) In the "Gene" tab of the application, individual genes are examined. When users inquire about a specific gene, multiple-testing correction of the p-value does not apply.

(2) In the "Tissue" tab, using adjusted p-values and a threshold of 0.05 yielded very few differentially expressed genes, limiting the utility of Peaks. Our objective therein is not to assess the significance of alterations in individual genes but to provide a metric for global alterations within a tissue. We then determine significance based on the False Discovery Rate (FDR), using the p-values as a nominal metric of gene expression alterations.

To avoid using the concept of “differential expression”, commonly linked to significance, we now refer to 'altered genes' in both the manuscript and the app. For clarity and to align with voyAGER's role as a hypothesis-generation tool, we define 'altered genes' as those with non-adjusted p-values < 0.01 or < 0.05 , as discriminated in the Methods section.

(3.3) In the gene-centric analyses section, authors should provide a full list of donor conditions and a summary table of conditions as supplementary.

We appreciate the suggestion and we have now included a reference that directs readers to those data, alternatively to including this information as an additional supplementary table. We would like to emphasise that the web app includes information on donor conditions we hypothesise to affect gene expression.

3.4) The tissue-specific assessment section has poor sub-titles. Every title has to contain information.

We agree and revised the sub-titles to more accurately reflect the information conveyed in each corresponding section.

(3.5) I have an issue understanding the meaning of NES from GSEA in the tissue-specific assessment section. The authors performed GSEA for the DEGs against the background genes ordered by tstatistics (from positive to negative) calculated from the linear model. I understand the p-value was two-tailed, which means that both positive and negative NES are meaningful as they represent up-regulated expression direction (positive coefficient) and down-regulated expression direction (negative coefficient) with age, respectively, within a window. However, in the GSEA section of Methods, authors were not fully elaborate on this directionality but stated, "The NES for each pathway was used in subsequent analyses as a metric of its over- or downrepresentation in the Peak". The authors should clearly elaborate on how to interpret the NES from their results.

We added the following paragraph to the manuscript's Methods section, in order to clarify the NES' directionality:

"We extracted the GSEA normalised enrichment score (NES), which represents the degree to which a certain gene set is overrepresented at the extreme ends of the ranked list of genes. A positive NES corresponds to the gene set's overrepresentation amongst up-regulated genes within the age window, whereas a negative NES signifies its overrepresentation amongst down-regulated genes. The NES for each pathway was used in subsequent analyses as a metric of its up- or down-regulation in the Peak."

(3.6) In the Modules of co-expressed genes section, the authors did not explain how or why they selected the four tissues: brain, skeletal muscle, heart (left ventricle), and whole blood. This should be elaborated on.

We apologise for not providing a detailed explanation for this selection. As the 'Modules of coexpressed genes' section was primarily intended as a proof of concept, we opted to include tissues for which we had a substantial number of samples available and availability of comprehensive cell type signatures, those being the tissues that met such criteria. Nonetheless, as the diversity of cell type signatures increases (e.g., through the increasing availability of scRNA-seq datasets), we plan to encompass a wider range of tissues in the near future. However, as this task is time-demanding and in order to avoid a substantial delay in the release of voyAGER, we opted to approach this issue in the next version of the App and included a dedicated issue in the projects' GitHub repository so that users can share their preferences of the next tissues to include.

We also added a brief sentence in this regard to the Methods section of the manuscript:

"The four tissues (Brain - Cortex, Muscle - Skeletal, Heart - Left Ventricle, and Whole Blood) covered by the Module section of voyAGER were selected due to their relatively high sample

sizes and availability of comprehensive cell type signatures. The increasing availability of human tissue scRNA-seq datasets (e.g., through the Human Cell Atlas) will allow future updates of voyAGER to encompass a wider range of tissues.”

(3.7) In the modules of the co-expressed genes section, the authors did not provide an explanation of the "diseases-manual" sub-tab of the "Pathway" tab of the voyAGER tool. It would be helpful for readers to understand how the candidate disease list was prepared and what the results represent.

We greatly appreciate the reviewer's feedback, and in response, we have restructured the 'Modules of co-expressed genes' method section to provide a more comprehensive explanation of the 'diseases' sub-section. To clarify, we obtained a curated set of diseases and their associated genes from DisGeNET v.7.0. We assessed the enrichment of modules in relation to these diseases through two methods: a manual approach utilising Fisher's tests (i.e. comparing the genes of a given module with the genes associated with a given disease) and another through use of the disgenet2r package, employing the function `disease_enrichment`. Significance of these enrichments were determined by adjusting p-values using the Benjamini-Hochberg correction.

(3.8) Most figures have low resolutions, and their fonts are too small to read.

As already mentioned in issue 2.10, we have recreated all of the images with better resolution to enhance legibility. We also exported such figures in PDF, which we attach to this revision.

(3.9) Authors used GTEx V7, which is not latest version. Although researchers have developed a huge amount of pipelines and tools for their research, most of them were neglected without a single update. I am sure many users, including myself, would appreciate it if the authors kept updating the database with GTEx V8 for the future version of the database.

We express our gratitude to the reviewer for their valuable suggestion, and, as already explained in issue 2.1, we have incorporated GTEx V8 into voyAGER.

(3.10) I would like to have an option for downloading the results as a whole for gene, tissue, and coexpressed genes. This would be a great option for secondary analysis by users.

The implementation of such feature would be a time-demanding endeavour that would delay the release of voyAGER, and we therefore chose not to perform it for this version. However, we agree that it would be a good resource for secondary analyses and acknowledge the possibility of adding this feature in the future. For now, voyAGER allows the user to download all plots and corresponding data.

(3.11) How the orders of tissues in the heatmaps (both gene and tissue section) were determined? Did the authors apply hierarchical clustering? If not, I would recommend the authors perform the hierarchical clustering and add it to display the heatmap display.

We apologise for the oversight in explaining the process behind determining the order of tissues. To clarify, we employed hierarchical clustering to establish the tissue order for visualisation within the app. Although the reviewer suggested adding a dendrogram to illustrate this clustering, we decided against it. The reason for such is that including a dendrogram, while informative, is not essential for the app's primary purpose.

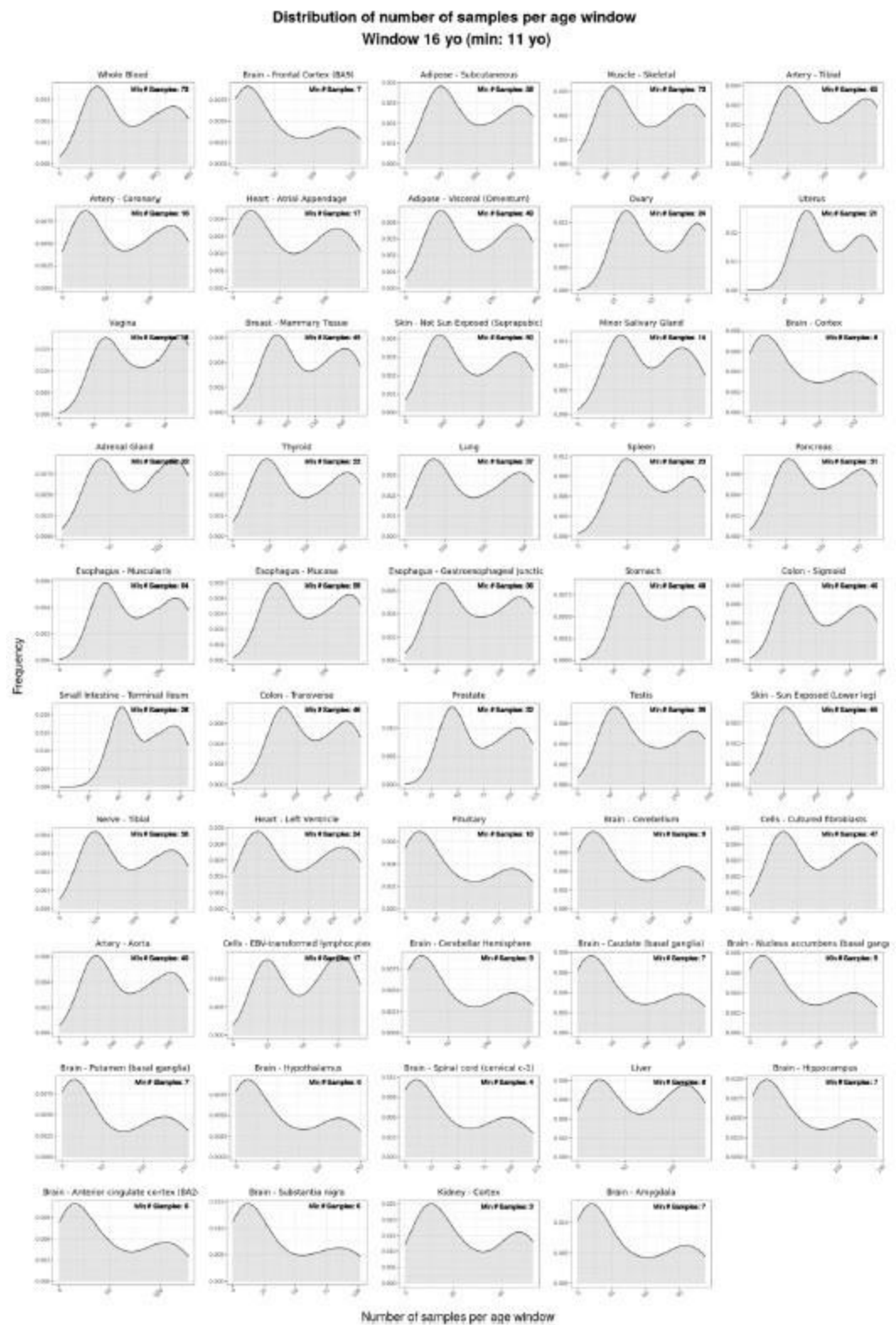
(3.12) I understand that this is a vast amount of work, but I hope that the authors can expand the coexpressed module analysis to include other tissues in the future version of the database.

Knowing what co-expressed genes in line with aging are and their pathway and disease enrichments across tissues would be highly informative, and I'm sure many users, including myself, would greatly appreciate it.

We express our gratitude to the reviewer for the valuable suggestion and for acknowledging the extensive effort required to incorporate new tissues into the module section. We completely agree that understanding co-expressed genes across the aging process is of significant value, and we are committed to the ongoing inclusion of additional tissues. As already stated in issue 3.6, comprehensive list of tissues slated for integration in future voyAGER versions is readily available on voyAGER's GitHub repository.

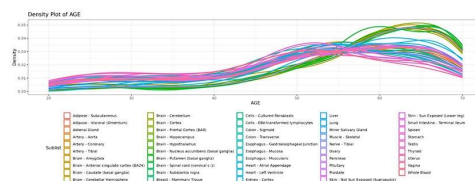
Author response image 1.

Density plots (“smoothed” histograms) of the distribution of numbers of samples per moving age window for the ShARP-LM pipeline, categorised by tissue. The numerical value within each rectangle represents the minimum number of samples observed across all age windows for that particular tissue.



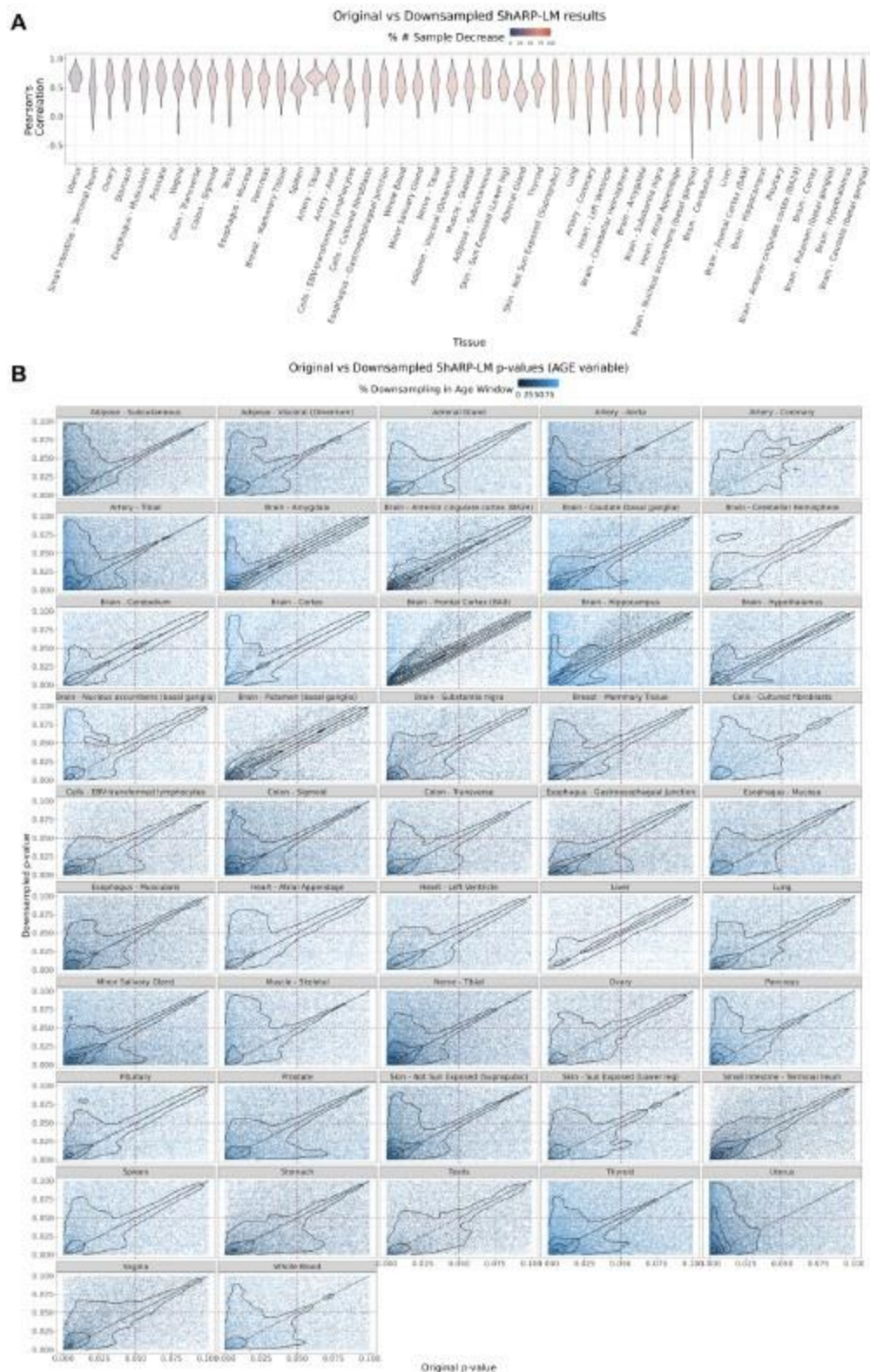
Author response image 2.

Density lines (“smoothed” histograms) of the distribution of the age of donors per tissue. As depicted in the chart, there are more samples for older ages, particularly of brain tissues.



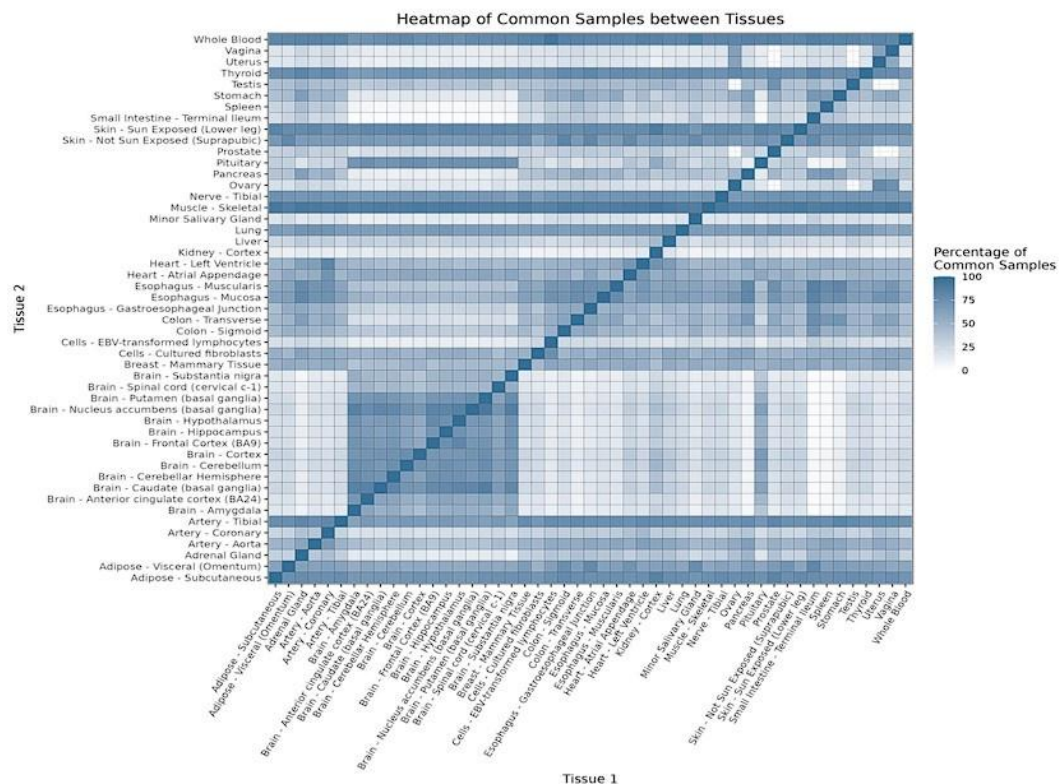
Author response image 3.

Effect of downsampling in ShARP-LM results. A – Per tissue violin plots of gene-wide distributions of Pearson’s correlation coefficients between original and downsampled logFC values for the Age variable across age windows, with tissues coloured by and ordered by increasing percentage of downsampling-associated reduction in the number of samples. B – Density scatter plots of comparison of associated original and downsampled p-values for each tissue, coloured by the downsampling percentage in each age window, highlighting the low range of p-values (from 0 to 0.1). Despite changes in logFC with downsampling, a considerable correlation in significance is maintained, although downsampling naturally results in a loss of statistical power, evident by the shift of points towards the first quadrant (dashed lines: p-value = 0.05).



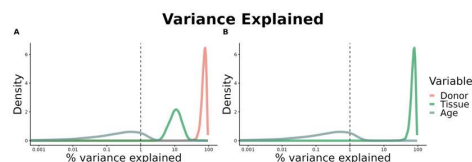
Author response image 4.

Heatmap depicting the percentage of common donors between pairs of tissues. A given square illustrates the percentage of all samples of tissue in the x axis (Tissue 1) that is in common with the tissue in the y axis (Tissue 2)



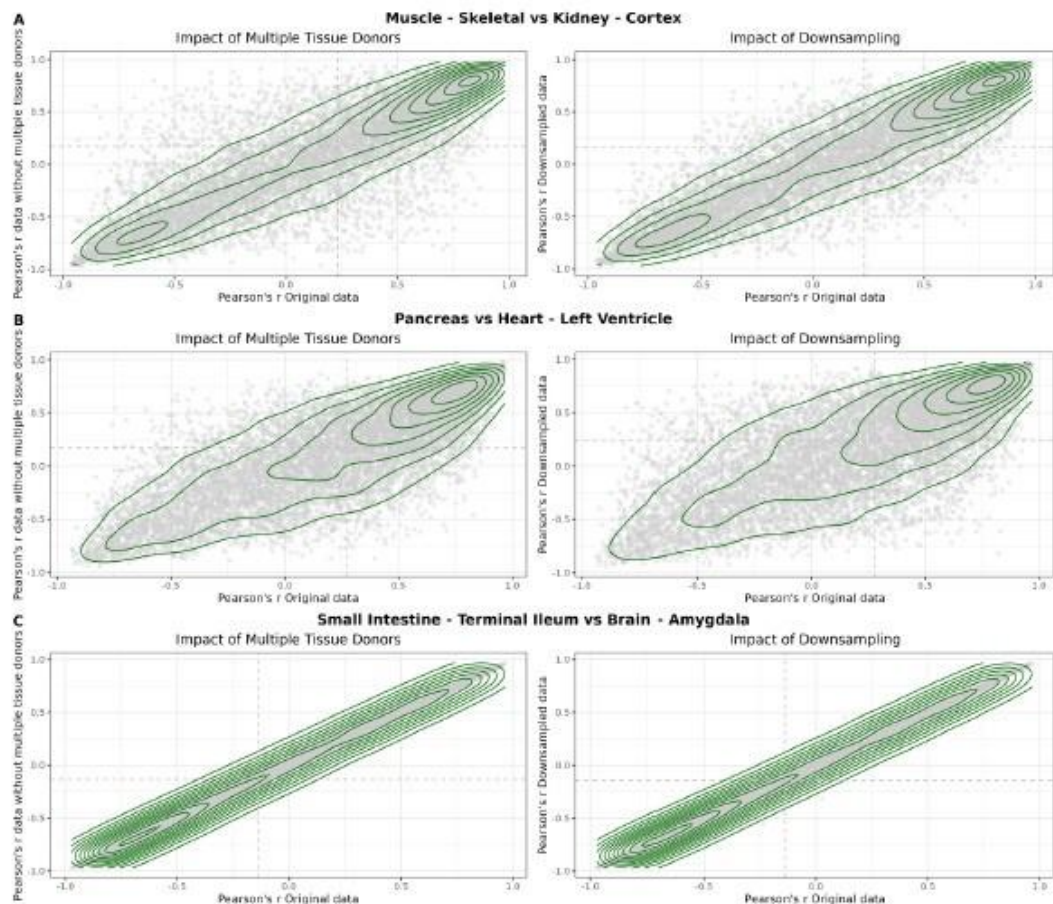
Author response image 5.

Assessment of the relative contributions of different sources to the dataset's variance. A - tissue accounts for approximately 90% of the total variance, while donor contributes around 10%; age has a minimal impact (1%), likely due to the relative subtlety of its effects on gene expression and to the tissue specificity of ageing dynamics. B - Removal of the donor variable does not transfer variance to age, suggesting limited confounding between the two variables.



Author response image 6.

Impact of the relative proportion of common donors on gene expression correlation between tissue pairs. Panels A, B, and C showcase the tissue pairs with the highest (Muscle Skeletal / Kidney Cortex), median (Pancreas / Heart Left Ventricle), and lowest (Small Intestine / Brain Amygdala) percentages of common donors, respectively. The left panels illustrate gene-by-gene Pearson's correlations of gene expression between the two tissues, comparing the scenarios with (x-axis) and without (y-axis) the removal of common donors. The right panels depict the same comparisons, but with random downsampling (y-axis) in both tissues based on the proportions defined for common donor removal. The depicted examples show that the outcomes are comparable when removing common donors or employing random downsampling.



Author response image 7.

Comparison of the impacts of removing common donor samples and random downsampling across tissue pairs. The heatmap is coloured based on whether the removal of common donors has a greater (red) or lesser impact (blue) than random downsampling. The values depicted in the heatmap, denoted as the Impact of Common Donors (ICD), are computed for each tissue pair. This calculation involves several steps: first, by determining the absolute difference in Pearson's correlation for each gene's mean expression within each age window from the ShARP-LM pipeline, between the original data and the subset of data without common donors (DiffWoCD) or with random downsampling (DiffRD). Subsequently, the medians of DiffWoCD and DiffRD are computed, and the difference between these median values provides the ICD for each tissue pair. Due to the unidirectional nature of correlation (i.e., the results for tissue 1 vs tissue 2 mirror those for tissue 2 vs tissue 1), the resulting matrix is triangular in form. Grey tiles denote NA values, i.e., where the tissue-tissue comparison does not have a meaning, namely self-self and between sex-specific tissues. Top right insert: density line ("smoothed" histogram) of all ICD values.

